
Breaking the Exploration Bottleneck: Rubric-Scaffolded Reinforcement Learning for Open-Ended LLM Reasoning

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Abstract

Recent advances in Large Language Models (LLMs) have underscored the potential of Reinforcement Learning (RL) to facilitate the emergence of reasoning capabilities. Despite the encouraging results, a fundamental dilemma persists as RL improvement relies on learning from high-quality samples, yet the exploration for such samples remains bounded by the inherent limitations of LLMs. This, in effect, creates an undesirable cycle in which *what cannot be explored cannot be learned*. In this work, we propose Rubric-Scaffolded Reinforcement Learning (RuscaRL), a novel instructional scaffolding framework designed to break the exploration bottleneck for open-ended reasoning. Specifically, RuscaRL introduces checklist-style rubrics as (1) *explicit scaffolding* for exploration during rollout generation, where different rubrics are provided as external guidance within task instructions to steer diverse high-quality responses. This guidance is gradually decayed over time, encouraging the model to internalize the underlying reasoning patterns; (2) *verifiable rewards* for exploitation during model training, where we can obtain robust LLM-as-a-Judge scores using rubrics as references, enabling effective RL on open-ended reasoning tasks. Extensive experiments demonstrate the superiority of the proposed RuscaRL across various benchmarks, effectively expanding reasoning boundaries under the Best-of-N evaluation.

1. Introduction

Large Language Models (LLMs) have demonstrated tremendous potential over a wide spectrum of complex reasoning tasks, including legal analysis (Choi et al., 2021; Lai et al., 2024; Fei et al., 2023; Trautmann et al., 2022), robotic manipulation (Driess et al., 2023; Zitkovich et al., 2023; Firoozi et al., 2025; Yao et al., 2023c), and real-world software development (Chen, 2021; Fan et al., 2023; Hou et al., 2024). However, enabling LLMs to reason reliably across open-ended, interactive, and safety-critical settings remains a significant challenge (Zhao et al., 2023; Huang & Chang, 2022; Liu et al., 2025d; Ying et al., 2026; 2025; Liang et al., 2025; Wang et al., 2025). To address this, recent breakthroughs in Reinforcement Learning with Verifiable Rewards (RLVR), as exemplified by DeepSeek-R1 (Guo et al., 2025), have proven that leveraging verifiable rewards as feedback signals can successfully facilitate the emergence of sophisticated reasoning capabilities in LLMs (Lambert et al., 2024; Yang et al., 2025; Kimi et al., 2025).

Despite the encouraging results, conventional RLVR tends to be more applicable to domains with objectively verifiable answers. For instance, in areas such as mathematical proof (Ren et al., 2025; Chen et al., 2025) and code generation (Qwen, 2025; Le et al., 2022), correctness can be explicitly determined through formal proof verification or automated unit tests. In these contexts, the reward signal is clear and well-aligned with the task objective, allowing RLVR to effectively guide models toward correct solutions. Unfortunately, many real-world tasks like medical consultation (Lin et al., 2025; Singhal et al., 2023; Zhang et al., 2023) and creative writing (Wu et al., 2025c; Franceschelli & Musolesi, 2024) are inherently open-ended. These tasks often require multidimensional evaluation and lack a single, verifiable ground-truth answer. To tackle this problem, several recent works (Arora et al., 2025; Kimi et al., 2025; Gunjal et al., 2025; Dou et al., 2025; Liu et al., 2025a) have explored rubric-based evaluation that decomposes desirable responses into checklist-style criteria (e.g., factuality, coherence, completeness) (Ho et al., 2024). By leveraging LLM-as-a-Judge to score each criterion and aggregating results into scalar rewards, rubrics provide more stable and

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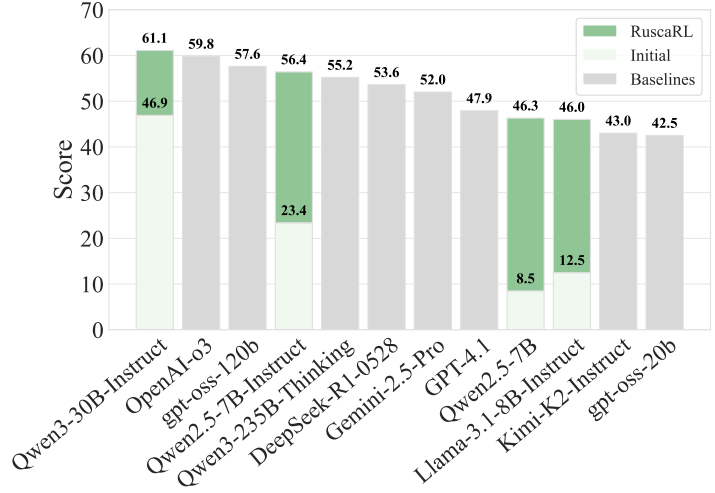
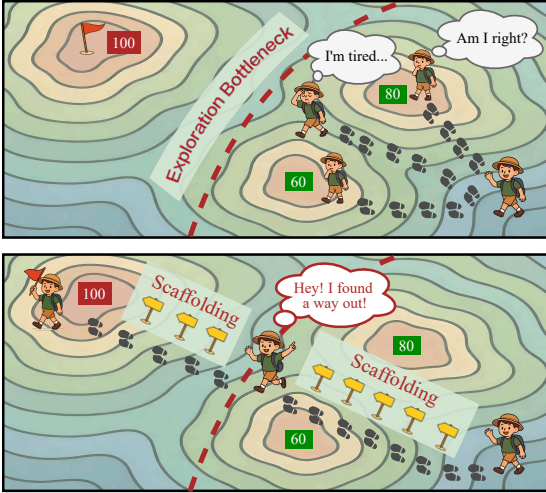


Figure 1. (Left) A conceptual illustration of exploration bottleneck and scaffolding. (Right) Performance comparison of different LLMs on HealthBench-500. Note that Qwen3-30B-Instruct denotes Qwen3-30B-A3B-Instruct-2507, and Qwen3-235B-Thinking denotes Qwen3-235B-A22B-Thinking-2507.

reliable feedback signals suitable for RLVR in open-ended domains.

Nevertheless, as shown in the left side of Figure 1, a fundamental exploration bottleneck remains as RL requires high-quality samples to improve, yet exploration for such samples remains bounded by the inherent capabilities of LLMs (Yue et al., 2025; Wu et al., 2025a; Liu et al., 2025c; Dong et al., 2025). This creates an inevitable loop where *the inability to explore restricts the ability to learn*. A growing line of work has attempted to enhance exploration in RLVR for LLMs (Liu et al., 2025c;b; Dong et al., 2025; Zheng et al., 2025; Lei et al., 2025; Li et al., 2025; Cheng et al., 2025). However, these methods largely bias the policy distribution toward high-reward responses already supported by base models, rather than truly expanding its reasoning boundaries (Wu et al., 2025a). Worse still, RL itself has a natural tendency to narrow the exploration space: policy entropy gradually collapses during training, causing the model to converge toward a limited set of reasoning trajectories (Zhao et al., 2025; Yue et al., 2025; Wu et al., 2025a; Yu et al., 2025; Liu et al., 2025c). This, in turn, undermines the potential of RLVR to explore more diverse and higher-quality solutions.

In this work, we introduce Rubric-Scaffolded Reinforcement Learning, termed as RuscaRL, which employs a novel instructional scaffolding framework to break the exploration bottleneck of RLVR. Technically, RuscaRL leverages rubrics in two complementary ways: (1) *Explicit scaffolding during rollout generation*. For each instruction, RuscaRL generates a group of candidate responses by using rubrics as external guidance. We propose intra-group scaffolding differentiation to provide varying levels of rubrics within

each group, enabling diverse and high-quality responses. To further internalize underlying reasoning patterns, we use inter-step scaffolding decay to gradually remove the scaffolding over training, thereby minimizing reliance on external guidance. (2) *Verifiable rewards during model training*. The model responses are assessed based on multiple criteria derived from rubrics. For each criterion, we employ a Grader LLM that performs a binary evaluation (*i.e.*, true or false), determining whether the response satisfies that specific requirement. The outcomes are then combined through aggregation to obtain a robust reward signal, facilitating effective RL across different general tasks. Our main contributions are summarized as follows:

- We introduce *instructional scaffolding* as a novel paradigm in RLVR for LLMs, which pioneers the integration of external guidance within task instructions to improve rollout diversity and quality, thereby enabling more efficient exploration during RL.
- We propose Rubric-Scaffolded Reinforcement Learning (RuscaRL), an innovative RLVR framework designed to break the exploration bottleneck, integrating checklist-style rubrics as both explicit scaffolding for *exploration* and verifiable rewards for *exploitation*.
- Extensive experiments demonstrate that RuscaRL yields highly competitive results compared to strong baseline methods. Notably, RuscaRL enables small LLMs (*e.g.*, Qwen3-30B-A3B-Instruct-2507) to achieve performance on par with leading LLMs (*e.g.*, OpenAI-o3) on HealthBench-500, as shown in the right side of Figure 1.

2. Related Works

LLM Reasoning. While early methods like prompt engineering (Wei et al., 2022; Kojima et al., 2022; Zhou et al., 2023a; Yao et al., 2023b;a) and supervised fine-tuning (Ouyang et al., 2022) have yielded encouraging results, their reliance on task-specific prompts or extensive labeled data limits their scalability and cross-domain generalization (Stiennon et al., 2020; Pornprasit & Tantithamthavorn, 2024; Zhang et al., 2024b; Gao et al., 2023). Recent works have found that using more test-time computation (Snell et al., 2024; Zhang et al., 2024a; Zuo et al., 2025; Zhang et al., 2026) can improve the reasoning performance of LLMs. More recently, RLVR (Lambert et al., 2024; OpenAI, 2025a; Guo et al., 2025) has emerged as a promising paradigm for training LLMs to solve verifiable problems, showing strong reasoning improvements in domains like math and coding (Guo et al., 2025; Qwen, 2025; Lambert et al., 2024; OpenAI, 2025a; Fang et al., 2025; Zhang et al., 2025b). However, it faces an exploration bottleneck (Wu et al., 2025a; Yue et al., 2025; Liu et al., 2025c) and is difficult to extend to general tasks where correctness is hard to verify (Gunjal et al., 2025; Kimi et al., 2025).

Rubric-based Methods. Rubrics are structured evaluation frameworks that decompose complex assessment tasks into specific, verifiable criteria. To address general task evaluation, rubric-based evaluation approaches have emerged across medical (Arora et al., 2025), code (Pathak et al., 2025), and other domains (Fan et al., 2025; Galvan-Sosa et al., 2025; Winata et al., 2025; Li et al., 2026). Building upon these frameworks, researchers apply rubrics as reward signals in RL (Kimi et al., 2025; Liu et al., 2025a; Gunjal et al., 2025), using LLMs as graders to provide fine-grained feedback for tasks lacking ground truth. This approach has shown promising results across agentic search (Liu et al., 2025a; Shao et al., 2025; Kimi et al., 2025), instruction following (Viswanathan et al., 2025; Huang et al., 2025), and medical consultation (Gunjal et al., 2025; Dou et al., 2025).

Exploration in RL for LLMs. Existing RL methods face insufficient exploration in complex reasoning tasks, with policies trapped in local optima and reasoning boundaries collapsing (Wu et al., 2025a; Yue et al., 2025; Liu et al., 2025c). Current solutions include prolonged training (Liu et al., 2025c;b), entropy-based exploration (Dong et al., 2025; Zheng et al., 2025; Lei et al., 2025; Li et al., 2025; Cheng et al., 2025), and external guidance (Zhang et al., 2025a; RRV et al., 2025), but these approaches fail to break the exploration bottleneck because they either explore within the initial policy distribution or provide only coarse directional signals without structured intermediate guidance. In contrast, RuscaRL provides rubrics as explicit scaffolding, guiding trajectories with verifiable criteria while enabling scaffolding decay for pattern internalization.

3. Preliminary

RL Algorithms for LLMs. We adopt Group Relative Policy Optimization (GRPO) (Shao et al., 2024) as our core RL algorithm for training language models with rubric-based rewards. Unlike Proximal Policy Optimization (PPO) (Schulman et al., 2017), GRPO eliminates the need for a value model by using group-based advantage estimation. For each instruction $q \sim \mathcal{D}$, where $\mathcal{D}$ denotes the distribution over the training dataset, GRPO samples a group of G responses $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$ and optimizes the policy π_{θ} by maximizing the following objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(\rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (1)$$

where o_i is a response sampled from the old policy $\pi_{\theta_{\text{old}}}$ given instruction q , t denotes the token position within response o_i , $\rho_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ is the token-level importance ratio between the current policy and the previous policy, and ϵ is the clipping coefficient (Schulman et al., 2017). The group-relative advantage is computed as:

$$\hat{A}_i = \frac{r_i - \text{mean}(\{r_j\}_{j=1}^G)}{\text{std}(\{r_j\}_{j=1}^G)}, \quad (2)$$

where r_i is the reward for response o_i , and the advantage is normalized using the mean and standard deviation of the G sampled rewards within each group.

4. Methodology

To address the exploration bottleneck problem, we propose the novel RuscaRL framework, as illustrated in Figure 2. RuscaRL effectively leverages rubrics in two complementary ways: (1) *Explicit scaffolding during rollout generation*, where the model generates candidate responses using rubrics as external guidance with intra-group differentiation and inter-step decay (Section 4.2); (2) *Verifiable rewards during model training*, where model responses are assessed based on multiple criteria derived from rubrics through binary evaluation and aggregation (Section 4.3). In the following, we first introduce the basic concept of rubrics, and then detail these two core components.

4.1. Rubric-based Evaluation System

Evaluating complex and open-ended tasks is inherently challenging, as responses often differ in structure, style, and content, making it difficult for rule-based evaluation to provide reliable assessments. To address this gap, recent work (Arora et al., 2025) has proposed rubric-based evaluation, which specifies fine-grained, multi-dimensional criteria

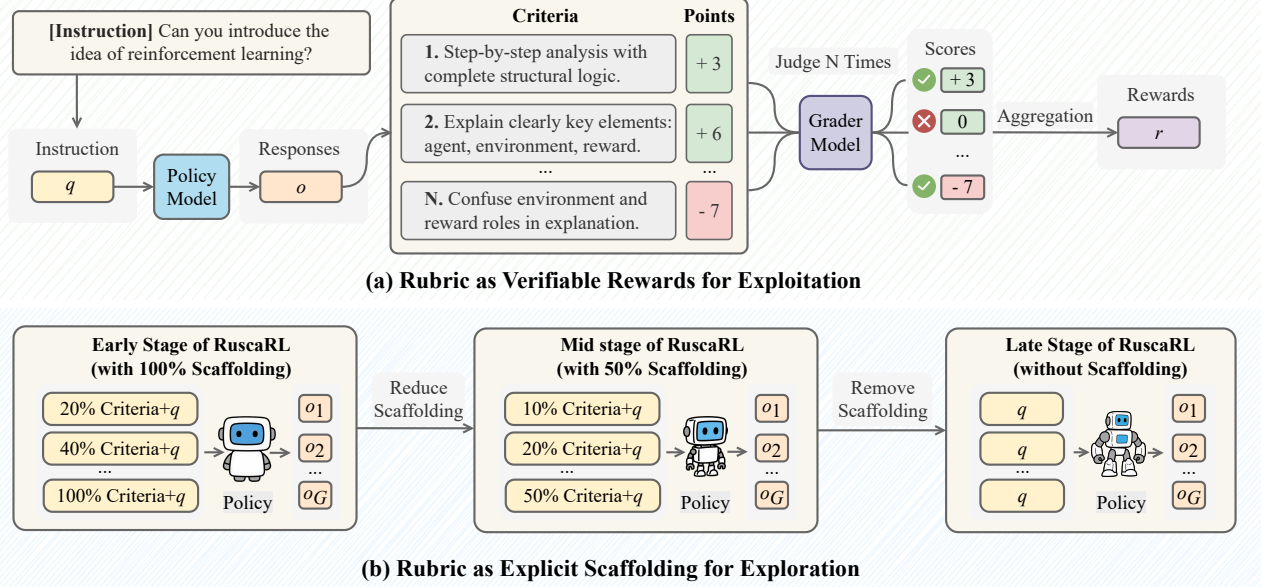


Figure 2. Overview of the RuscaRL framework. (a) Rubric as Verifiable Rewards for Exploitation: checklist-style rubric criteria with associated points are used by an LLM-based grader to perform criterion-wise binary judgments and aggregate the resulting scores into a scalar reward for each response. (b) Rubric as Explicit Scaffolding for Exploration: the same rubrics are injected into instructions as external guidance during rollout generation, with intra-group scaffolding differentiation providing different levels of criteria to samples within a group and inter-step scaffolding decay gradually reducing the amount of scaffolding over training.

that can be applied at scale. This design combines the objectivity of automatic metrics with the principled guidance of structured standards, yielding robust scores that better capture response quality, coherence, and completeness.

Formally, a rubric $\mathcal{R} = \{c_1, c_2, \dots, c_N\}$ is defined as a set of N verifiable criteria. Each criterion c_i is specified by a clear description and corresponding points p_i indicating its contribution to the overall evaluation. We define the point vector as $\mathbf{p} = [p_1, p_2, \dots, p_N]$. For example, given the instruction “Can you introduce the idea of reinforcement learning?”, criteria may include: “step-by-step analysis with complete structural logic” (+3 points), “explain key elements: agent, environment, reward” (+6 points), and negative items like “confuse environment and reward roles in explanation” (-7 points). Points are added or subtracted depending on whether each criterion is satisfied.

Given an instruction q and its corresponding rubric $\mathcal{R}$ both sampled from the data distribution $\mathcal{D}$, and a model response o generated through the policy model $\pi_\theta(o|q)$, we first construct a judge prompt for each criterion c_i by combining the instruction q , response o , and criterion c_i . The judge prompt template for the grader is provided in Appendix D.1. For a single criterion evaluation, the grader function $\mathcal{G}$ implemented by a LLM (Zheng et al., 2023b; Gu et al., 2024) takes the judge prompt as input and outputs a binary decision $b_i = \mathcal{G}(q, o, c_i) \in \{0, 1\}$ indicating whether criterion c_i is satisfied (true or false). Extending this to the full

rubric, the grader evaluates all criteria and produces a binary indicator vector $\mathbf{b} = \mathcal{G}(q, o, \mathcal{R}) = [b_1, b_2, \dots, b_N]$ where each b_i represents the satisfaction of criterion c_i . The final score vector is obtained by element-wise multiplication: $\mathbf{s} = \mathbf{b} \odot \mathbf{p} = [b_1 p_1, b_2 p_2, \dots, b_N p_N]$, providing fine-grained score across all specified criteria. We also compute the total possible score $S_{total} = \sum_{j=1}^M p_j$ where M is the number of positive-point criteria, which will be used for normalization in the reward calculation.

4.2. Rubric-based Scaffolding for RL Exploration

During RL training on complex reasoning tasks, models often fail to sustain effective exploration (Wu et al., 2025a; Yue et al., 2025; Liu et al., 2025c): initial randomness quickly diminishes, policy entropy collapses, and the model prematurely converges to suboptimal reasoning patterns. This collapse severely limits the discovery of diverse and high-quality solution trajectories.

To mitigate this issue, we draw inspiration from instructional scaffolding theory in educational psychology (Vygotsky, 1978). According to Vygotsky’s Zone of Proximal Development, when learners’ capabilities are insufficient, they benefit from structured guidance to bridge the gap between current ability and target performance. As competence grows, this scaffolding should be gradually withdrawn to foster independent problem-solving (Wood et al., 1976).

Building on this insight, we design a rubric-based scaffolding mechanism that provides varying numbers of rubric criteria as explicit guidance throughout the training process, helping models gradually learn to generate high-quality responses. Specifically, our rubric-based scaffolding mechanism augments the original policy function by adding a subset of rubric criteria $\mathcal{R}_S$ as additional guidance, representing the policy as $\pi_\theta(o|q, \mathcal{R}_S)$. The specific prompt template for incorporating scaffolding is detailed in Appendix D.2. Additionally, we design a two-dimensional control mechanism to determine the rubrics scaffolding ratio λ_S , and then sample criteria from the complete rubric set $\mathcal{R}$ to form $\mathcal{R}_S$, *i.e.*, $|\mathcal{R}_S| = \text{round}(\lambda_S \times |\mathcal{R}|)$. We instantiate this mechanism in two dimensions: intra-group Scaffolding differentiation and inter-step Scaffolding decay.

Intra-Group Scaffolding Differentiation. In RL algorithms with multi-sampling, such as GRPO, computing group-relative advantages (Eq. 2) requires response diversity to avoid collapse into homogeneous samples. To this end, we assign different levels of rubric scaffolding within each group, encouraging both guided and independent exploration. Concretely, we define a group-level ratio vector $\lambda_{group} = [\lambda_1, \lambda_2, \dots, \lambda_G]$ where $\lambda_i = \frac{G-i}{G-1}$ for the i -th sample in the group of size G . This linear differentiation ensures that some samples benefit from stronger scaffolding while others are deliberately exposed to weaker guidance, thereby enhancing intra-group diversity.

Inter-Step Scaffolding Decay. Inspired by instructional scaffolding theory, we gradually reduce guidance as the model develops independent learning strategies using a sigmoid function $\lambda_{step}(t) = \frac{1}{1+e^{\alpha(t-t_0)}}$, where t is the current training progress ($t \in [0, 1]$), t_0 is the midpoint, and α controls the steepness of decay. This mechanism prevents overfitting to external guidance by creating an adaptive learning environment where the model initially benefits from guidance to overcome the exploration bottleneck, then gradually transitions to independent reasoning as capabilities mature.

Integrated Scaffolding Mechanism. Finally, we combine the intra-group differentiation and the inter-step decay into a unified scaffolding ratio vector:

$$\lambda_S = \lambda_{step}(t) \times \lambda_{group} = [\lambda_{S,1}, \lambda_{S,2}, \dots, \lambda_{S,G}], \quad (3)$$

where $\lambda_{S,i} = \lambda_{step}(t) \times \lambda_i = \frac{1}{1+e^{\alpha(t-t_0)}} \times \frac{G-i}{G-1}$ represents the scaffolding ratio for the i -th sample in the group. This integrated mechanism simultaneously promotes response diversity within each group and adaptively reduces reliance on scaffolding across training steps, jointly addressing the problems of homogeneity and overfitting.

4.3. Rubric-based Reward for RL Exploitation

To provide robust and reliable reward signals for open-ended reasoning tasks, we design the following rubric-

based reward functions. The multi-dimensional score vector $s = [s_1, s_2, \dots, s_N]$ obtained from the evaluation system is then aggregated into a final scalar reward by directly summing all criterion scores and normalizing by the total possible score computed in Section 4.1:

$$S = \frac{\sum_{i=1}^N s_i}{S_{total}}, \quad (4)$$

where S represents the final score, s_i is the score of the i -th criterion, and S_{total} is the total possible score of all positive-point criteria computed in Section 4.1. This calculation method results in scores that fall within the interval $[0, 1]$ in most cases, with occasional negative scores possible. We directly adopt this rubric-based score S as our reward: $r_i = S_i$, where r_i is the reward for the i -th response. This approach enables application to open-ended tasks without ground truth answers while providing more robust assessment than holistic LLM scoring.

Once the rubric-based rewards are obtained, we employ them to train the policy model using RL algorithms. The training process follows the policy gradient framework, where the model learns to maximize the expected reward. Algorithm 1 in Appendix A outlines the complete training procedure. Additionally, to help the model better internalize underlying reasoning patterns, the log probability computation in training is based on $\pi_\theta(o_{i,t}|q, o_{i,<t})$ rather than $\pi_\theta(o_{i,t}|q, \mathcal{R}_S, o_{i,<t})$. For detailed analysis on importance sampling, see Appendix C.6.

5. Experiments

To demonstrate the effectiveness of the proposed RuscaRL method, we conduct experiments across multiple benchmarks spanning medical, writing, instruction following, and STEM domains. Our experiments seek to answer the following questions: (1) Does RuscaRL demonstrate consistent effectiveness across different models and tasks, and how does it compare against existing fine-tuning methods? (Section 5.2 and Appendix C.1, C.2, C.3, C.4) (2) How does RuscaRL break the exploration bottleneck in RL for LLM reasoning? (Section 5.3 and Appendix C.5) (3) What is the impact of different components in the rubric-based scaffolding mechanism? (Section 5.4)

5.1. Experimental Setups

Models and Training Settings. We use several backbone models from different series and with varying parameter scales for our experiments, including both instruction-tuned models and base models from the Qwen2.5 series (Yang et al., 2024), the Qwen3 series (Yang et al., 2025), and the Llama-3 series (Dubey et al., 2024). All these models are trained using the GRPO algorithm on the verl framework (Sheng et al., 2025). Detailed training settings are

Table 1. Main results comparison with baseline models across different benchmarks. The best results in each box are highlighted in **bold**.

Model	Medical				Writing		Instruction Following	
	HealthBench-500	LLMEval-Med	MedQA	MedMCQA	WritingBench	Creative Writing	IFEVAL	IFBench
Closed-source and Open-source Models								
OpenAI-o3	59.8	78.9	96.0	84.7	77.7	81.4	91.6	67.8
GPT-4.1	47.9	71.2	92.4	80.0	69.0	79.0	87.0	37.2
gpt-oss-20b	42.5	68.8	85.6	68.1	66.6	39.4	83.5	48.7
Rubicon-Preview	50.8	73.2	85.1	70.7	73.0	66.4	82.4	33.4
Our Models								
Qwen3-30B-A3B-Instruct	46.9	71.5	84.2	71.3	78.1	74.4	83.0	31.9
+ RuscaRL	61.1 (+14.2)	73.2 (+1.7)	84.8 (+0.6)	71.9 (+0.6)	79.2 (+1.1)	74.3 (-0.1)	84.5 (+1.5)	32.1 (+0.2)
Qwen3-30B-A3B-Base	11.2	43.1	73.6	65.1	36.9	35.8	39.0	13.3
+ RuscaRL	48.4 (+37.2)	60.9 (+17.8)	71.3 (-2.3)	65.4 (+0.3)	59.5 (+22.6)	46.0 (+10.2)	76.3 (+37.3)	30.3 (+17.0)
Qwen2.5-7B-Instruct	23.4	48.0	61.8	56.3	45.2	37.4	71.0	26.8
+ RuscaRL	56.4 (+33.0)	65.3 (+17.3)	63.5 (+1.7)	56.5 (+0.2)	56.1 (+10.9)	38.6 (+1.2)	75.3 (+4.3)	31.0 (+4.2)
Qwen2.5-7B	8.5	28.2	55.3	55.0	23.8	30.3	32.0	14.5
+ RuscaRL	46.3 (+37.8)	47.9 (+19.7)	58.2 (+2.9)	55.6 (+0.6)	46.0 (+22.2)	34.8 (+4.5)	56.2 (+24.2)	25.9 (+11.4)
Llama-3.1-8B-Instruct	12.5	30.1	66.8	58.0	36.7	44.5	72.6	22.6
+ RuscaRL	46.0 (+33.5)	46.2 (+16.1)	70.7 (+3.9)	60.7 (+2.7)	52.7 (+16.0)	54.2 (+9.7)	79.7 (+7.1)	31.1 (+8.5)
Llama-3.1-8B	0.0	9.1	36.9	35.9	13.0	26.3	18.1	11.6
+ RuscaRL	25.8 (+25.8)	29.6 (+20.5)	49.7 (+12.8)	45.4 (+9.5)	35.7 (+22.7)	33.3 (+7.0)	55.6 (+37.5)	21.4 (+9.8)

provided in Appendix B.1.

Evaluation Benchmarks. We use HealthBench-500, a randomly selected subset of 500 samples from HealthBench (Arora et al., 2025), as our held-out evaluation set. Additionally, we evaluate on other medical benchmarks including LLMEval-Med (Zhang et al., 2025c), MedQA (Jin et al., 2021), and MedMCQA (Pal et al., 2022). For the Writing domain, we use WritingBench (Wu et al., 2025c) and Creative Writing v3 (Paech, 2025) benchmarks. For the Instruction Following domain, we use IFEVAL (Zhou et al., 2023b) and IFBench (Pyatkin et al., 2025) benchmarks.

Baselines. We compare RuscaRL against several representative baseline methods: (1) Rubric-based RL: A simple GRPO baseline using rubric scores as rewards, corresponding to Rubric as Verifiable Rewards for Exploitation without scaffolding mechanisms. (2) MeRF (Zhang et al., 2025a): Following the MeRF paradigm of injecting reward specifications as in-context motivation, we include all rubrics in the prompt as scaffolding throughout training, without intra-group differentiation or inter-step decay. (3) RL-Plus (Dong et al., 2025): A hybrid-policy RL method that combats capability boundary collapse through mixture-policy importance sampling and exploration-oriented advantage shaping, encouraging continued discovery of correct but under-represented reasoning paths. (4) Entropy-Augmented RL (EA-RL) (Cheng et al., 2025): An advantage-shaping approach that adds a stabilized entropy bonus to encourage exploratory reasoning and mitigate exploration collapse. (5) SFT: Fine-tuned on GPT-4.1 (OpenAI, 2025b) demonstrations generated with scaffolding support. (6) SFT + Rubric-based RL: A combination approach that first applies SFT and then applies rubric-based RL training.

5.2. Overall Performance

RuscaRL achieves consistent and notable gains across tasks and model scales, showcasing its effectiveness and broad generalization. Across medical, writing, and instruction-following tasks (Table 1), RuscaRL achieves significant gains over multiple initial models, with Qwen3-30B-A3B-Instruct on HealthBench-500 surpassing many leading models (e.g., OpenAI-o3). It is worth noting that our training data consists of open-ended tasks, whereas MedQA and MedMCQA are closed-ended multiple-choice benchmarks. Our improvements on these closed-ended benchmarks are marginal and are included only to demonstrate cross-task generalization. Notably, RuscaRL is particularly effective on instruction-tuned models and provides larger benefits for weaker models, such as Llama-3.1-8B-Instruct. This advantage stems from our scaffolding approach, which leverages the model’s existing instruction-following ability to elicit higher-quality and more diverse responses, explaining why RuscaRL is especially well-suited for training on models with strong instruction-following capacities. More detailed results about performance across different model series and scales are provided in Appendix C.1, which further demonstrates the robustness and broad applicability of our approach. Additionally, we explore the effects of mixing training data from different domains in Appendix C.3.

RuscaRL consistently outperforms Rubric-based methods across tasks. As shown in Table 2, in the **direct RL** setting, RuscaRL achieves the best performance on most medical, writing, and instruction-following tasks, delivering larger and more stable gains than Rubric-based RL and MeRF (e.g., 56.4 vs. 52.0 and 36.8 accuracy on HealthBench-500 with Qwen2.5-7B-Instruct). Compared to exploration-enhanced methods RL-Plus and EA-RL, Rus-

Table 2. Performance comparison of baseline methods across different benchmarks. The best results in each box are highlighted in **bold**.

Method	Medical				Writing		Instruction Following	
	HealthBench-500	LLMEval-Med	MedQA	MedMCQA	WritingBench	Creative Writing	IFEVAL	IFBench
Qwen2.5-7B-Instruct								
Initial	23.4	48.0	61.8	56.3	45.2	37.4	71.0	26.8
Rubric-based RL	52.0 (+28.6)	56.5 (+8.5)	62.3 (+0.5)	56.3 (+0.0)	53.7 (+8.5)	38.8 (+1.4)	75.1 (+4.1)	29.3 (+2.5)
MeRF	36.8 (+13.4)	56.1 (+8.1)	57.9 (-3.9)	52.4 (-3.9)	45.9 (+0.7)	38.3 (+0.9)	71.9 (+0.9)	28.6 (+1.8)
RL-Plus	53.6 (+30.2)	58.2 (+10.2)	62.7 (+0.9)	56.4 (+0.1)	54.8 (+9.6)	38.5 (+1.1)	75.0 (+4.0)	29.8 (+3.0)
EA-RL	51.3 (+27.9)	57.8 (+9.8)	61.5 (-0.3)	56.5 (+0.2)	52.9 (+7.7)	39.2 (+1.8)	74.6 (+3.6)	28.7 (+1.9)
RuscaRL (Ours)	56.4 (+33.0)	65.3 (+17.3)	63.5 (+1.7)	56.5 (+0.2)	56.1 (+10.9)	38.6 (+1.2)	75.3 (+4.3)	31.0 (+4.2)
SFT	38.3 (+14.9)	52.6 (+4.6)	60.8 (-1.0)	57.3 (+1.0)	62.8 (+17.6)	45.3 (+7.9)	75.2 (+4.2)	25.2 (-1.6)
SFT + Rubric-based RL	55.5 (+32.1)	58.5 (+10.5)	59.7 (-2.1)	56.4 (+0.1)	66.7 (+21.5)	43.6 (+6.2)	82.1 (+11.1)	29.6 (+2.8)
SFT + RuscaRL (Ours)	56.9 (+33.5)	58.8 (+10.8)	61.6 (-0.2)	56.9 (+0.6)	67.0 (+21.8)	43.9 (+6.5)	82.5 (+11.5)	30.6 (+3.8)
Qwen2.5-7B								
Initial	8.5	28.2	55.3	55.0	23.8	30.3	32.0	14.5
Rubric-based RL	42.0 (+33.5)	46.5 (+18.3)	48.2 (-7.1)	49.9 (-5.1)	40.1 (+16.3)	33.8 (+3.5)	53.4 (+21.4)	25.5 (+11.0)
MeRF	21.7 (+13.2)	44.4 (+16.2)	60.3 (+5.0)	55.5 (+0.5)	43.4 (+19.6)	25.7 (-4.6)	52.3 (+20.3)	20.4 (+5.9)
RL-Plus	43.8 (+35.3)	47.2 (+19.0)	49.6 (-5.7)	50.8 (-4.2)	42.3 (+18.5)	34.2 (+3.9)	54.6 (+22.6)	25.7 (+11.2)
EA-RL	41.2 (+32.7)	45.8 (+17.6)	47.5 (-7.8)	51.2 (-3.8)	41.5 (+17.7)	33.2 (+2.9)	54.1 (+22.1)	24.8 (+10.3)
RuscaRL (Ours)	46.3 (+37.8)	47.9 (+19.7)	58.2 (+2.9)	55.6 (+0.6)	46.0 (+22.2)	34.8 (+4.5)	56.2 (+24.2)	25.9 (+11.4)
SFT	32.2 (+23.7)	40.0 (+11.8)	56.5 (+1.2)	54.4 (-0.6)	56.6 (+32.8)	42.5 (+12.2)	69.7 (+37.7)	20.8 (+6.3)
SFT + Rubric-based RL	36.5 (+28.0)	39.7 (+11.5)	57.1 (+1.8)	54.1 (-0.9)	57.4 (+33.6)	43.2 (+12.9)	71.6 (+39.6)	23.7 (+9.2)
SFT + RuscaRL (Ours)	35.4 (+26.9)	42.7 (+14.5)	58.2 (+2.9)	55.1 (+0.1)	57.7 (+33.9)	42.6 (+12.3)	72.0 (+40.0)	23.1 (+8.6)

caRL also demonstrates superior performance, indicating that our scaffolding mechanism provides more effective exploration guidance than entropy-based or hybrid-policy approaches. In the **SFT-then-RL** setting, both RuscaRL and Rubric-based RL achieve additional improvements over SFT, with RuscaRL yielding larger gains across most tasks, though the magnitude is smaller than in the direct RL setting. We argue that RuscaRL essentially leverages rubrics as prior knowledge to guide exploration, while SFT also serves to accelerate RL exploration (cold start). Since both mechanisms overlap in facilitating exploration, this may explain why the performance gap between RuscaRL and Rubric-based RL narrows under the SFT-then-RL setting.

5.3. Analysis

In this subsection, we conduct a detailed analysis of RuscaRL using Qwen2.5-7B-Instruct as the initial model and HealthBench as the training and evaluation dataset. In addition, we compare RuscaRL and Rubric-based RL in the following analysis. We use the Best-of-N¹ metric to reflect both the model’s reasoning boundaries at large N and sampling efficiency at small N. We further use novelty and diversity metrics to diagnose exploration behavior; their definitions and detailed statistics are provided in Appendix C.5.

RuscaRL substantially improves sampling efficiency and reasoning boundaries. As shown in Figure 3a, RuscaRL substantially improves single-sample quality at N=1, indicating that the scaffolding mechanism effectively enhances

the model’s reasoning stability. At N=2048, its performance ceiling surpasses both the initial model and Rubric-based RL, validating its advantage in expanding the reasoning boundary. Moreover, RuscaRL exhibits a steeper performance curve across N, meaning it can achieve the same performance with fewer samples.

RuscaRL also *produces highly novel responses that the initial model could barely generate*, showing that rubric scaffolding effectively breaks the exploration bottleneck and discovers new solutions. Figure 4a further shows that this advantage is consistent across novelty metrics, suggesting broader movement beyond the initial policy’s likely output region. Detailed importance-ratio statistics and high-novelty samples are provided in Appendix C.5.

RuscaRL achieves exploration-exploitation balance. As shown in Figure 3b, RuscaRL demonstrates a well-balanced exploration-exploitation trajectory: policy entropy first rises as the model explores diverse reasoning trajectories, then declines as it converges to high-quality patterns. In contrast, Rubric-based RL plateaus under continuous entropy decline. Validation accuracy (Figure 3c) consistently shows RuscaRL achieving the best performance throughout training, demonstrating sustained improvement without bottlenecks. The corresponding sigmoid scaffolding-decay schedule is shown in Figure 3d.

RuscaRL also *maintains higher training-time sampling diversity*, indicating that rubric scaffolding supports broader exploration before stable exploitation. Similar trends are observed in Self-BLEU and Semantic Distance. As shown in Figures 4b and 4c, RuscaRL rapidly improves diversity in the early training stages, then maintains a relatively sta-

¹For cost considerations in the Best-of-N evaluation, we use Qwen3-32B (non-thinking) as the Grader LLM.

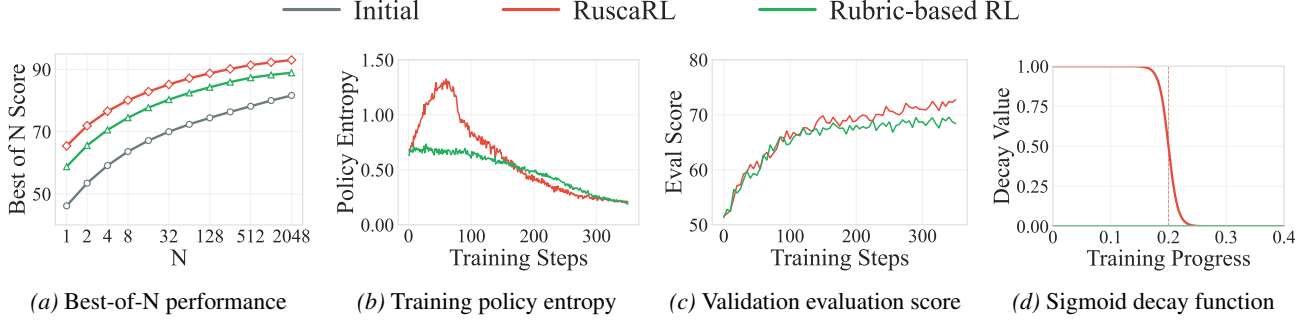


Figure 3. Best-of-N evaluation and training dynamics. We compare RuscaRL and Rubric-based RL across sampling efficiency, policy entropy, validation score, and the sigmoid decay schedule.

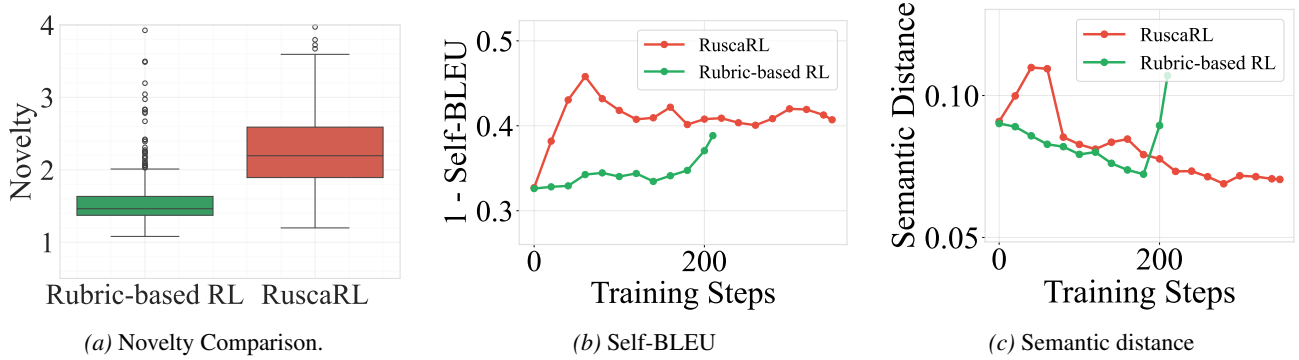


Figure 4. Novelty and diversity analysis. (a) Novelty Comparison. (b,c) Diversity comparison during training.

ble high diversity level with a gradual decline. In contrast, Rubric-based RL shows faster diversity collapse, especially on semantic distance. These results confirm that RuscaRL achieves effective exploration followed by stable exploitation.

5.4. Ablation Studies

Intra-group Differentiation Analysis. We first analyze different strategies for the intra-group control mechanism using Qwen2.5-7B-Instruct as the initial model and HealthBench as the training and evaluation dataset. Within individual sampling groups, we compare different rubric scaffolding differentiation patterns. These mechanisms are: **(1) Linear (Ours):** Linear differentiation pattern following our proposed formula $\frac{G-i}{G-1}$, providing different levels of rubric scaffolding to different samples within a single sampling group. **(2) Binary:** Binary differentiation patterns where N represents the number of samples with full rubric scaffolding within a single sampling group, including configurations such as no-scaffolding ($N=0$), half-scaffolding ($N=4$), and full-scaffolding ($N=8$).

As shown in Figure 5a, the linear differentiation strategy performs optimally in intra-group control, outperforming both no-scaffolding ($N=0$) and fully scaffolded ($N=8$) binary settings. This result indicates that the gain does not

come from persistently emphasizing heavily scaffolded trajectories, but from differentiating scaffolding levels within each sampling group to improve diversity, which works synergistically with multi-sampling algorithms like GRPO.

Inter-step Decay Analysis. We then analyze different decay functions for inter-step control during training. We formally define the base scaffolding intensity of inter-step control as $f(t)$, where t is the normalized training progress ($t \in [0, 1]$). We compare the following decay functions: **(1) Sigmoid (Ours):** S-shaped decay function $f(t) = \frac{1}{1+e^{\alpha(t-t_0)}}$, where parameter α controls the steepness of decay and t_0 controls the midpoint of decay, achieving smooth nonlinear transitions. **(2) Constant:** Constant control $f(t) = 1$, maintaining constant full scaffolding throughout training. **(3) Linear:** Linear decay function $f(t) = 1 - t$, achieving uniform linear decrease. **(4) Power (n):** Power decay function $f(t) = (1 - t)^n$, where n controls the curvature of decay, including various power configurations.

As shown in Figure 5b, the sigmoid decay function achieves the best performance among all decay strategies, while constant full scaffolding performs much worse. This indicates that persistent reliance on scaffolded prompts is not the source of improvement. Instead, RuscaRL benefits from a decayed scaffolding schedule: the sigmoid function provides sufficient guidance in the early training stage to

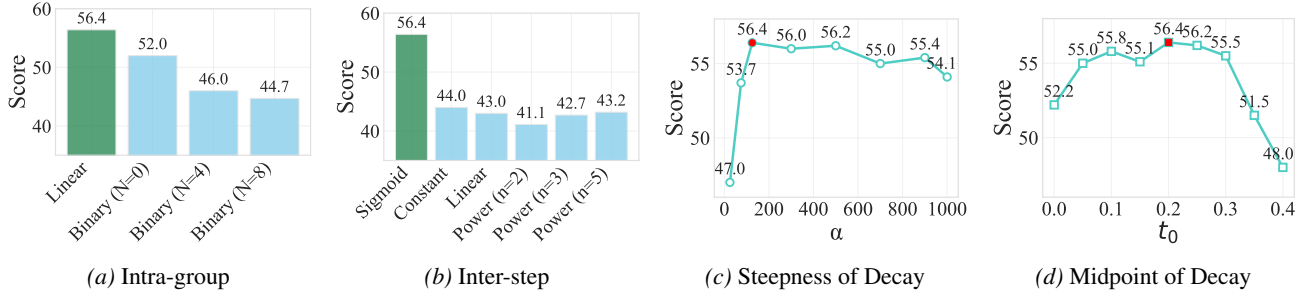


Figure 5. Ablation studies on RuscaRL framework components. (a) Intra-group differentiation strategies comparison; (b) Inter-step decay functions comparison; (c) Sigmoid parameter steepness of decay α sensitivity analysis with fixed $t_0 = 0.2$; (d) Sigmoid parameter midpoint of decay t_0 sensitivity analysis with fixed $\alpha = 125$.

support exploration, and then gradually reduces external guidance so that the model can internalize useful reasoning patterns. In contrast, linear and power decay strategies perform worse, likely because their prolonged scaffolding can cause the model to overfit to external guidance rather than focus on the original instruction.

Based on the superior performance of the sigmoid function, we further analyze the effects of both parameter dimensions (speed α and midpoint t_0) using Qwen2.5-7B-Instruct as the initial model and HealthBench as the training and evaluation dataset. Figures 5c and 5d demonstrate the performance differences across various sigmoid parameter configurations, ultimately determining the optimal configuration as $\alpha = 125$, $t_0 = 0.2$. (1) Removing scaffolding too fast (large α) prevents the model from adapting to rapid scaffolding changes, easily causing training instability; while removing scaffolding too slowly (small α) leads to incomplete early-stage scaffolding, failing to fully stimulate the model’s exploration capability, and prolonged retention of scaffolding in later stages also causes overfitting issues. (2) Starting decay too early (small t_0) leads to insufficient scaffolding support, causing the model to lack necessary guidance in early training stages; while starting decay too late (large t_0) causes the model to over-rely on scaffolding, ultimately resulting in overfitting. Notably, RuscaRL maintains strong performance and consistently outperforms Rubric-based RL across a wide range of hyperparameters, with degradation only at extreme values.

6. Conclusion and Discussion

In this work, we apply instructional scaffolding theory to RL for LLMs, and introduce RuscaRL, a novel instructional scaffolding framework that breaks the exploration bottleneck for open-ended LLM reasoning tasks. RuscaRL leverages checklist-style rubrics through scaffolding mechanisms that provide gradually decaying external guidance to encourage internalization, while the rubric-based reward function enables robust and fine-grained evaluation for ef-

fective RL training on general tasks. Extensive experiments demonstrate that RuscaRL consistently outperforms strong baseline methods and achieves competitive results compared with leading models. Furthermore, the model fine-tuned with RuscaRL produces highly novel responses that the initial model could barely generate, effectively expanding the reasoning boundary. The success of RuscaRL underscores the urgent need for broader community investment in constructing open, diverse, and domain-rich rubric datasets. Our future work includes developing pipelines for high-quality rubric data production, exploring rubric-based natural language feedback strategies, and investigating applications to multi-modal tasks and agent systems.

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Impact Statement

This paper presents work whose goal is to advance reinforcement learning methods for improving open-ended reasoning capabilities in large language models. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

Arora, R. K., Wei, J., Hicks, R. S., Bowman, P., Quiñonero-Candela, J., Tsimplouras, F., Sharman, M., Shah, M., Vallone, A., Beutel, A., et al. Healthbench: Evaluating large language models towards improved human health. *arXiv preprint arXiv:2505.08775*, 2025.

- Bai, Y., Zhang, J., Lv, X., Zheng, L., Zhu, S., Hou, L., Dong, Y., Tang, J., and Li, J. Longwriter: Unleashing 10,000+ word generation from long context llms. *arXiv preprint arXiv:2408.07055*, 2024.
- Chen, L., Gu, J., Huang, L., Huang, W., Jiang, Z., Jie, A., Jin, X., Jin, X., Li, C., Ma, K., et al. Seed-prover: Deep and broad reasoning for automated theorem proving. *arXiv preprint arXiv:2507.23726*, 2025.
- Chen, M. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- Cheng, D., Huang, S., Zhu, X., Dai, B., Zhao, W. X., Zhang, Z., and Wei, F. Reasoning with exploration: An entropy perspective. *arXiv preprint arXiv:2506.14758*, 2025.
- Choi, J. H., Hickman, K. E., Monahan, A. B., and Schwarcz, D. Chatgpt goes to law school. *J. Legal Educ.*, 71:387, 2021.
- Dong, Y., Jiang, X., Tao, Y., Liu, H., Zhang, K., Mou, L., Cao, R., Ma, Y., Chen, J., Li, B., et al. RL-plus: Countering capability boundary collapse of llms in reinforcement learning with hybrid-policy optimization. *arXiv preprint arXiv:2508.00222*, 2025.
- Dou, C., Liu, C., Yang, F., Li, F., Jia, J., Chen, M., Ju, Q., Wang, S., Dang, S., Li, T., et al. Baichuan-m2: Scaling medical capability with large verifier system. *arXiv preprint arXiv:2509.02208*, 2025.
- Driess, D., Xia, F., Sajjadi, M. S. M., Lynch, C., Chowdhery, A., Ichter, B., Wahid, A., Tompson, J., Vuong, Q., Yu, T., Huang, W., Chebotar, Y., Sermanet, P., Duckworth, D., Levine, S., Vanhoucke, V., Hausman, K., Toussaint, M., Greff, K., Zeng, A., Mordatch, I., and Florence, P. Palm-e: an embodied multimodal language model. In *ICML*, 2023.
- Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Yang, A., Fan, A., et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Fan, A., Gokkaya, B., Harman, M., Lyubarskiy, M., Sen-gupta, S., Yoo, S., and Zhang, J. M. Large language models for software engineering: Survey and open problems. In *ICSE-FoSE*, pp. 31–53, 2023.
- Fan, Z., Wang, W., Wu, X., and Zhang, D. Sedareval: Automated evaluation using self-adaptive rubrics. *arXiv preprint arXiv:2501.15595*, 2025.
- Fang, W., Liu, S., Zhou, Y., Zhang, K., Zheng, T., Chen, K., Song, M., and Tao, D. Serl: Self-play reinforcement learning for large language models with limited data. In *Advances in Neural Information Processing Systems*, 2025.
- Fei, Z., Shen, X., Zhu, D., Zhou, F., Han, Z., Zhang, S., Chen, K., Shen, Z., and Ge, J. Lawbench: Benchmarking legal knowledge of large language models. *arXiv preprint arXiv:2309.16289*, 2023.
- Firoozi, R., Tucker, J., Tian, S., Majumdar, A., Sun, J., Liu, W., Zhu, Y., Song, S., Kapoor, A., Hausman, K., et al. Foundation models in robotics: Applications, challenges, and the future. *IJRR*, 44(5):701–739, 2025.
- Franceschelli, G. and Musolesi, M. On the creativity of large language models. *AI & SOCIETY*, pp. 1–11, 2024.
- Galvan-Sosa, D., Gaudeau, G., Kavumba, P., Li, Y., Yuan, Z., Sakaguchi, K., Buttery, P., et al. Rubrik’s cube: Testing a new rubric for evaluating explanations on the cube dataset. *arXiv preprint arXiv:2503.23899*, 2025.
- Gao, L., Madaan, A., Zhou, S., Alon, U., Liu, P., Yang, Y., Callan, J., and Neubig, G. Pal: Program-aided language models. In *ICML*, pp. 10764–10799, 2023.
- Google. Gemini 2.5 pro: Best for coding and highly complex tasks, 2025. URL <https://deepmind.google/models/gemini/pro/>.
- Gu, J., Jiang, X., Shi, Z., Tan, H., Zhai, X., Xu, C., Li, W., Shen, Y., Ma, S., Liu, H., et al. A survey on llm-as-a-judge. *arXiv preprint arXiv:2411.15594*, 2024.
- Gunjal, A., Wang, A., Lau, E., Nath, V., Liu, B., and Hendryx, S. Rubrics as rewards: Reinforcement learning beyond verifiable domains. *arXiv preprint arXiv:2507.17746*, 2025.
- Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. In *ICLR*, 2021a.
- Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., and Steinhardt, J. Measuring mathematical problem solving with the math dataset. In *NeurIPS*, volume 1, 2021b.
- Ho, Z. Y., Liang, S., Zhang, S., Zhan, Y., and Tao, D. Novo: Norm voting off hallucinations with attention heads in large language models. *arXiv preprint arXiv:2410.08970*, 2024.
- Hou, X., Zhao, Y., Liu, Y., Yang, Z., Wang, K., Li, L., Luo, X., Lo, D., Grundy, J., and Wang, H. Large language models for software engineering: A systematic literature review. *ACM TOSEM*, 33(8):1–79, 2024.

- Huang, J. and Chang, K. C.-C. Towards reasoning in large language models: A survey. *arXiv preprint arXiv:2212.10403*, 2022.
- Huang, Z., Zhuang, Y., Lu, G., Qin, Z., Xu, H., Zhao, T., Peng, R., Hu, J., Shen, Z., Hu, X., et al. Reinforcement learning with rubric anchors. *arXiv preprint arXiv:2508.12790*, 2025.
- Jin, D., Pan, E., Oufattole, N., Weng, W.-H., Fang, H., and Szolovits, P. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.
- Kimi, Bai, Y., Bao, Y., Chen, G., Chen, J., Chen, N., Chen, R., Chen, Y., Chen, Y., Chen, Y., et al. Kimi k2: Open agentic intelligence. *arXiv preprint arXiv:2507.20534*, 2025.
- Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., and Iwasawa, Y. Large language models are zero-shot reasoners. In *NeurIPS*, volume 35, pp. 22199–22213, 2022.
- Lai, J., Gan, W., Wu, J., Qi, Z., and Yu, P. S. Large language models in law: A survey. *AI Open*, 5:181–196, 2024.
- Lambert, N., Morrison, J., Pyatkin, V., Huang, S., Ivison, H., Brahman, F., Miranda, L. J. V., Liu, A., Dziri, N., Lyu, S., et al. Tulu 3: Pushing frontiers in open language model post-training. *arXiv preprint arXiv:2411.15124*, 2024.
- Le, H., Wang, Y., Gotmare, A. D., Savarese, S., and Hoi, S. C. H. Coderl: Mastering code generation through pretrained models and deep reinforcement learning. In *NeurIPS*, volume 35, pp. 21314–21328, 2022.
- Lei, S., Cheng, Z., Jia, K., and Tao, D. Revisiting llm reasoning via information bottleneck. *arXiv preprint arXiv:2507.18391*, 2025.
- Li, S., Zhao, J., Wei, M., Ren, H., Zhou, Y., Yang, J., Liu, S., Zhang, K., and Chen, W. Rubrichub: A comprehensive and highly discriminative rubric dataset via automated coarse-to-fine generation. *arXiv preprint arXiv:2601.08430*, 2026.
- Li, X., Callanan, E., Ghassel, A., and Zhu, X. Entropy-gated branching for efficient test-time reasoning. *arXiv preprint arXiv:2503.21961*, 2025.
- Liang, S., Fang, T., Liu, Z., Liu, A., Xiao, Y., He, J., Chang, E.-C., and Cao, X. Safemobile: Chain-level jailbreak detection and automated evaluation for multimodal mobile agents. *arXiv preprint arXiv:2507.00841*, 2025.
- Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and Cobbe, K. Let’s verify step by step. In *ICLR*, 2023.
- Lin, T., Zhang, W., Li, S., Yuan, Y., Yu, B., Li, H., He, W., Jiang, H., Li, M., Song, X., et al. Healthgpt: A medical large vision-language model for unifying comprehension and generation via heterogeneous knowledge adaptation. *arXiv preprint arXiv:2502.09838*, 2025.
- Liu, A., Mei, A., Lin, B., Xue, B., Wang, B., Xu, B., Wu, B., Zhang, B., Lin, C., Dong, C., et al. Deepseek-v3. 2: Pushing the frontier of open large language models. *arXiv preprint arXiv:2512.02556*, 2025a.
- Liu, M., Diao, S., Hu, J., Lu, X., Dong, X., Zhang, H., Bukharin, A., Zhang, S., Zeng, J., Sreedhar, M. N., et al. Scaling up rl: Unlocking diverse reasoning in llms via prolonged training. *arXiv preprint arXiv:2507.12507*, 2025b.
- Liu, M., Diao, S., Lu, X., Hu, J., Dong, X., Choi, Y., Kautz, J., and Dong, Y. Prorl: Prolonged reinforcement learning expands reasoning boundaries in large language models. *arXiv preprint arXiv:2505.24864*, 2025c.
- Liu, S., Fang, W., Hu, Z., Zhang, J., Zhou, Y., Zhang, K., Tu, R., Lin, T.-E., Huang, F., Song, M., and Tao, D. A survey of direct preference optimization. *arXiv preprint arXiv:2503.11701*, 2025d.
- Lu, D., Tan, X., Xu, R., Yao, T., Qu, C., Chu, W., Xu, Y., and Qi, Y. Scp-116k: A high-quality problem-solution dataset and a generalized pipeline for automated extraction in the higher education science domain. *arXiv preprint arXiv:2501.15587*, 2025.
- OpenAI. Introducing openai o3 and o4-mini, 2025a. URL <https://openai.com/index/introducing-o3-and-o4-mini/>.
- OpenAI. Introducing gpt-4.1 in the api, 2025b. URL <https://openai.com/index/gpt-4-1/>.
- OpenAI. Introducing gpt-oss, 2025c. URL <https://openai.com/index/introducing-gpt-oss/>.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. In *NeurIPS*, volume 35, pp. 27730–27744, 2022.
- Paech, S. J. Eq-bench creative writing benchmark v3. <https://github.com/EQ-bench/creative-writing-bench>, 2025.
- Pal, A., Umaphathi, L. K., and Sankarasubbu, M. Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *CHIL*, pp. 248–260, 2022.

- Papineni, K., Roukos, S., Ward, T., and Zhu, W.-J. Bleu: a method for automatic evaluation of machine translation. In *ACL*, pp. 311–318, 2002.
- Pathak, A., Gandhi, R., Uttam, V., Nakka, Y., Jindal, A. R., Ghosh, P., Ramamoorthy, A., Verma, S., Mittal, A., Ased, A., et al. Rubric is all you need: Enhancing llm-based code evaluation with question-specific rubrics. *arXiv preprint arXiv:2503.23989*, 2025.
- Pornprasit, C. and Tantithamthavorn, C. Fine-tuning and prompt engineering for large language models-based code review automation. *Information and Software Technology*, 175:107523, 2024.
- Pyatkin, V., Malik, S., Graf, V., Ivison, H., Huang, S., Dasigi, P., Lambert, N., and Hajishirzi, H. Generalizing verifiable instruction following. *arXiv preprint arXiv:2507.02833*, 2025.
- Qwen. Qwen3-coder: Agentic coding in the world, 2025. URL <https://qwenlm.github.io/blog/qwen3-coder/>.
- Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., Michael, J., and Bowman, S. R. GPQA: A graduate-level google-proof q&a benchmark. In *COLM*, 2024.
- Ren, Z., Shao, Z., Song, J., Xin, H., Wang, H., Zhao, W., Zhang, L., Fu, Z., Zhu, Q., Yang, D., et al. Deepseek-prover-v2: Advancing formal mathematical reasoning via reinforcement learning for subgoal decomposition. *arXiv preprint arXiv:2504.21801*, 2025.
- RRV, A., Dineen, J., Handa, D., Uddin, M. N., Parmar, M., Baral, C., and Zhou, B. Thinktuning: Instilling cognitive reflections without distillation. *arXiv preprint arXiv:2508.07616*, 2025.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Shao, R., Asai, A., Shen, S. Z., Ivison, H., Kishore, V., Zhuo, J., Zhao, X., Park, M., Finlayson, S. G., Sontag, D., et al. Dr tulul: Reinforcement learning with evolving rubrics for deep research. *arXiv preprint arXiv:2511.19399*, 2025.
- Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y., Wu, Y., et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Sheng, G., Zhang, C., Ye, Z., Wu, X., Zhang, W., Zhang, R., Peng, Y., Lin, H., and Wu, C. Hybridflow: A flexible and efficient rlhf framework. In *EuroSys*, pp. 1279–1297, 2025.
- Singhal, K., Azizi, S., Tu, T., Mahdavi, S. S., Wei, J., Chung, H. W., Scales, N., Tanwani, A., Cole-Lewis, H., Pfohl, S., et al. Large language models encode clinical knowledge. *Nature*, 620(7972):172–180, 2023.
- Snell, C., Lee, J., Xu, K., and Kumar, A. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.
- Stiennon, N., Ouyang, L., Wu, J., Ziegler, D., Lowe, R., Voss, C., Radford, A., Amodei, D., and Christiano, P. F. Learning to summarize with human feedback. In *NeurIPS*, volume 33, pp. 3008–3021, 2020.
- Trautmann, D., Petrova, A., and Schilder, F. Legal prompt engineering for multilingual legal judgement prediction. *arXiv preprint arXiv:2212.02199*, 2022.
- Viswanathan, V., Sun, Y., Ma, S., Kong, X., Cao, M., Neubig, G., and Wu, T. Checklists are better than reward models for aligning language models. *arXiv preprint arXiv:2507.18624*, 2025.
- Vygotsky, L. S. *Mind in society: The development of higher psychological processes*. Harvard university press, 1978.
- Wang, L., Ying, Z., Zhang, T., Liang, S., Hu, S., Zhang, M., Liu, A., and Liu, X. Manipulating multimodal agents via cross-modal prompt injection. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 10955–10964, 2025.
- Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., Jiang, Z., et al. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. In *NeurIPS*, volume 37, pp. 95266–95290, 2024.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, volume 35, pp. 24824–24837, 2022.
- Winata, G. I., Anugraha, D., Liu, E., Aji, A. F., Hung, S.-Y., Parashar, A., Irawan, P. A., Zhang, R., Yong, Z.-X., Cruz, J. C. B., et al. Datasheets aren’t enough: Datarubrics for automated quality metrics and accountability. *arXiv preprint arXiv:2506.01789*, 2025.
- Wood, D., Bruner, J. S., and Ross, G. The role of tutoring in problem solving. *JCPP*, 17(2):89–100, 1976.
- Wu, F., Xuan, W., Lu, X., Harchaoui, Z., and Choi, Y. The invisible leash: Why rlvr may not escape its origin. *arXiv preprint arXiv:2507.14843*, 2025a.

- Wu, Y., Bai, Y., Hu, Z., Lee, R. K.-W., and Li, J. Longwriter-zero: Mastering ultra-long text generation via reinforcement learning. *arXiv preprint arXiv:2506.18841*, 2025b.
- Wu, Y., Mei, J., Yan, M., Li, C., Lai, S., Ren, Y., Wang, Z., Zhang, J., Wu, M., Jin, Q., et al. Writingbench: A comprehensive benchmark for generative writing. *arXiv preprint arXiv:2503.05244*, 2025c.
- Xu, Z., Dai, X., Wei, S., Yin, S., and Hu, Y. Gspo: A graph substitution and parallelization joint optimization framework for dnn inference. In *DAC*, pp. 1–6, 2024.
- Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T., Cao, Y., and Narasimhan, K. Tree of thoughts: Deliberate problem solving with large language models. In *NeurIPS*, volume 36, pp. 11809–11822, 2023a.
- Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K. R., and Cao, Y. React: Synergizing reasoning and acting in language models. In *ICLR*, 2023b.
- Yao, X., Zhou, H., Mees, O., Meng, Y., Xiao, T., Bisk, Y., Oh, J., Johns, E., Shridhar, M., Shah, D., Thomason, J., Huang, K., Chai, J., Bing, Z., and Knoll, A. Bridging language and action: A survey of language-conditioned robot manipulation. *arXiv preprint arXiv:2312.10807*, 2023c.
- Ying, Z., Zhang, D., Jing, Z., Xiao, Y., Zou, Q., Liu, A., Liang, S., Zhang, X., Liu, X., and Tao, D. Reasoning-augmented conversation for multi-turn jailbreak attacks on large language models. *arXiv preprint arXiv:2502.11054*, 2025.
- Ying, Z., Liu, A., Liang, S., Huang, L., Guo, J., Zhou, W., Liu, X., and Tao, D. Safebench: A safety evaluation framework for multimodal large language models. *International Journal of Computer Vision*, 134(1):18, 2026.
- Yu, Q., Zhang, Z., Zhu, R., Yuan, Y., Zuo, X., Yue, Y., Dai, W., Fan, T., Liu, G., Liu, L., et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- Yue, Y., Chen, Z., Lu, R., Zhao, A., Wang, Z., Song, S., and Huang, G. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.
- Zhang, D., Zhoubian, S., Hu, Z., Yue, Y., Dong, Y., and Tang, J. Rest-mcts*: Llm self-training via process reward guided tree search. In *NeurIPS*, volume 37, pp. 64735–64772, 2024a.
- Zhang, H., Chen, J., Jiang, F., Yu, F., Chen, Z., Li, J., Chen, G., Wu, X., Zhang, Z., Xiao, Q., et al. Huatuoqpt, towards taming language model to be a doctor. *arXiv preprint arXiv:2305.15075*, 2023.
- Zhang, J., Ma, G., Liu, S., Wang, H., Huang, J., Lin, T.-E., Huang, F., Li, Y., and Tao, D. A simple ”motivation” can enhance reinforcement finetuning of large reasoning models. *arXiv preprint arXiv:2506.18485*, 2025a.
- Zhang, K., Yao, Q., Liu, S., Wang, Y., Lai, B., Ye, J., Song, M., and Tao, D. Consistent paths lead to truth: Self-rewarding reinforcement learning for llm reasoning. In *Advances in Neural Information Processing Systems*, 2025b.
- Zhang, K., YAO, Q., Lai, B., Huang, J., Fang, W., Tao, D., Song, M., and Liu, S. Incentivizing llm reasoning via reinforcement learning with functional monte carlo tree search. In *International Conference on Learning Representations*, 2026.
- Zhang, M., Shen, Y., Li, Z., Sha, H., Hu, B., Wang, Y., Huang, C., Liu, S., Tong, J., Jiang, C., et al. Llm eval-med: A real-world clinical benchmark for medical llms with physician validation. *arXiv preprint arXiv:2506.04078*, 2025c.
- Zhang, X., Talukdar, N., Vemulapalli, S., Ahn, S., Wang, J., Meng, H., Murtaza, S. M. B., Leshchiner, D., Dave, A. A., Joseph, D. F., et al. Comparison of prompt engineering and fine-tuning strategies in large language models in the classification of clinical notes. *AMIA Summits on Translational Science Proceedings*, 2024:478, 2024b.
- Zhang, Y., Li, M., Long, D., Zhang, X., Lin, H., Yang, B., Xie, P., Yang, A., Liu, D., Lin, J., et al. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025d.
- Zhao, R., Meterez, A., Kakade, S., Pehlevan, C., Jelassi, S., and Malach, E. Echo chamber: RL post-training amplifies behaviors learned in pretraining. *arXiv preprint arXiv:2504.07912*, 2025.
- Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.
- Zheng, C., Ke, P., Zhang, Z., and Huang, M. Click: Controllable text generation with sequence likelihood contrastive learning. In *Findings of ACL*, pp. 1022–1040, 2023a.

- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. Judging llm-as-a-judge with mt-bench and chatbot arena. In *NeurIPS*, volume 36, pp. 46595–46623, 2023b.
- Zheng, T., Xing, T., Gu, Q., Liang, T., Qu, X., Zhou, X., Li, Y., Wen, Z., Lin, C., Huang, W., et al. First return, entropy-eliciting explore. *arXiv preprint arXiv:2507.07017*, 2025.
- Zhou, D., Schärli, N., Hou, L., Wei, J., Scales, N., Wang, X., Schuurmans, D., Cui, C., Bousquet, O., Le, Q., et al. Least-to-most prompting enables complex reasoning in large language models. In *ICLR*, 2023a.
- Zhou, J., Lu, T., Mishra, S., Brahma, S., Basu, S., Luan, Y., Zhou, D., and Hou, L. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023b.
- Zhu, Y., Lu, S., Zheng, L., Guo, J., Zhang, W., Wang, J., and Yu, Y. Taxygen: A benchmarking platform for text generation models. In *SIGIR*, pp. 1097–1100, 2018.
- Zitkovich, B., Yu, T., Xu, S., Xu, P., Xiao, T., Xia, F., Wu, J., Wohlhart, P., Welker, S., Wahid, A., et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *CoRL*, pp. 2165–2183, 2023.
- Zuo, Y., Zhang, K., Sheng, L., Qu, S., Cui, G., Zhu, X., Li, H., Zhang, Y., Long, X., Hua, E., et al. Ttrl: Test-time reinforcement learning. *arXiv preprint arXiv:2504.16084*, 2025.

A. Algorithm Pseudocode

Algorithm 1 provides the complete pseudocode for our RuscaRL training procedure, illustrating the key components including intra-group scaffolding differentiation, inter-step scaffolding decay, and rubric-based reward computation.

Algorithm 1 RuscaRL Algorithm

```

1: Input: Policy model  $\pi_\theta$ , data distribution  $\mathcal{D}$ , grader model  $\mathcal{G}$ 
2: Initialize: Reference policy  $\pi_{ref} \leftarrow \pi_\theta$ 
3: for each training iteration  $t$  do
4:   for each  $(q, \mathcal{R}) \sim \mathcal{D}$  do
5:     Compute scaffolding ratio vector:  $\lambda_S = \lambda_{step}(t) \times \lambda_{group}$ 
6:     for  $i = 1$  to  $G$  do
7:       Sample rubric subset  $\mathcal{R}_S \subset \mathcal{R}$  based on  $\lambda_{S,i}$ 
8:       Generate response:  $o_i \sim \pi_\theta(\cdot | q, \mathcal{R}_S)$ 
9:     end for
10:    for each response  $o_i$  do
11:      Evaluate with grader:  $\mathbf{b}_i = \mathcal{G}(q, o_i, \mathcal{R})$ 
12:      Compute score vector:  $\mathbf{s} = \mathbf{b} \odot \mathbf{p}$ 
13:      Compute reward:  $S = \frac{\sum_{i=1}^N s_i}{S_{total}}$ 
14:    end for
15:    Compute advantages based on rewards
16:    Update policy model  $\pi_\theta$ 
17:  end for
18:  Update scaffolding step ratio:  $\lambda_{step}(t)$ 
19: end for
20: Return: Trained policy  $\pi_\theta$ 
    
```

B. Detailed Experimental Settings

B.1. Detailed Training Settings

Initial Models. We conducted training on models across different series and parameter scales, including the Qwen2.5 series (Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-7B, Qwen2.5-32B-Instruct, and Qwen2.5-32B), the Qwen3 series (Qwen3-4B-Instruct-2507, Qwen3-4B-Base, Qwen3-30B-A3B-Instruct-2507, and Qwen3-30B-A3B-Base), and the Llama-3 series (Llama-3.1-8B-Instruct, Llama-3.1-8B, and Llama-3.2-3B-Instruct).

Training Datasets. For the medical domain, we use the remaining 4500 samples from HealthBench after excluding HealthBench-500. For the other domains, we generate HealthBench-like rubrics data by calling GPT-4.1 (OpenAI, 2025b) with specific prompts detailed in Appendix D.3. For the writing domain, we combine LongWriter-6k (Bai et al., 2024) and LongWriter-Zero-RLData (Wu et al., 2025b) datasets. For the instruction following domain, we use IF-multi-constraints-upto5 (Pyatkin et al., 2025) dataset. For the STEM domain, we use SCP-116K (Lu et al., 2025) and MATH training datasets (Level 3-5) (Hendrycks et al., 2021b).

Training Configurations. This section provides detailed training configurations, as shown in Table 3. All models share identical hyperparameters except for the t_0 parameter in the sigmoid decay function and the sampling parameters. For the t_0 parameter, Qwen3-30B-A3B-Instruct and Qwen3-30B-A3B-Base use $t_0 = 0.1$, Llama-3.1-8B-Instruct and Llama-3.1-8B use $t_0 = 0.15$, Llama-3.2-3B-Instruct uses $t_0 = 0.3$, and the remaining models (Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-7B, Qwen2.5-32B-Instruct, Qwen2.5-32B, Qwen3-4B-Instruct-2507 and Qwen3-4B-Base) use $t_0 = 0.2$. For sampling parameters, we use a conservative decoding strategy (Temperature=0.7, Top-P=0.8, Top-K=20) for most models. However, we observed that Qwen2.5-7B-Instruct exhibits rapid policy entropy collapse under this configuration. Therefore, for Qwen2.5-7B-Instruct, we use Temperature=1.0 with Top-P and Top-K disabled.

Table 3. RuscaRL training configuration (Qwen2.5-7B-Instruct).

Category	Configuration
RuscaRL	Inter-step Scaffolding Decay: Step Sigmoid ($\alpha = 125$, $t_0 = 0.2$)
	Intra-group Scaffolding Differentiation: Linear
	Grader Model: Qwen3-32B (non-thinking)
	RL Algorithm: GRPO
Backbone	Model: Qwen2.5-7B-Instruct
Sampling	Temperature: 1.0
	Top-P: 1.0, Top-K: -1
	Rollout Samples per Prompt: 8
	Max Response Length: 4096
Training	Optimizer: Adam
	Learning Rate: 1×10^{-6} (Constant)
	Training Batch Size: 64
	Mini Batch Size: 32
	KL Loss Coefficient: 1×10^{-3}
	Entropy Coefficient: 0
Hardware	Epochs: 5
	GPUs: $8 \times \text{H200}$

B.2. Detailed Evaluation Settings

For medical benchmarks (HealthBench-500 and LLMEval-Med), we employ GPT-4.1 as the judge model. For writing benchmarks (WritingBench and Creative Writing v3), we employ Claude-Sonnet-4 as the judge model. Our generation parameters are set to Temperature=0.7, Top-P=0.8, and Top-K=20 across all evaluations. The maximum output length is configured as 4096 tokens for non-writing tasks and 16000 tokens for writing tasks. For IFEVAL and IFBench, we report the prompt-level strict-accuracy metric. We report the average of three runs for all evaluation benchmarks. All scores are converted to a percentage scale for reporting.

We also compare against other models, including closed-source models (OpenAI-o3 (OpenAI, 2025a), GPT-4.1 (OpenAI, 2025b), Gemini-2.5-Pro (Google, 2025)) and open-source models (DeepSeek-R1-0528 (Guo et al., 2025), Qwen3-235B-Thinking-2507 (Yang et al., 2025), Kimi-K2-Instruct (Kimi et al., 2025), gpt-oss-120b, gpt-oss-20b (OpenAI, 2025c), Rubicon-Preview (Huang et al., 2025)), on HealthBench-500 (Figure 1), to demonstrate the competitiveness of our approach.

C. Additional Experimental Analysis

C.1. Performance Across Different Models

Table 4 shows the performance comparison between initial model performance and RuscaRL-enhanced performance, demonstrating improvements across different model series and scales.

To further compare against the simple GRPO baseline on stronger base models, Table 5 reports results on HealthBench-500 and LLMEval-Med for two Qwen3 instruction-tuned models. RuscaRL consistently outperforms Rubric-based RL on both models, indicating that rubric scaffolding remains effective even when the initial model already has stronger capabilities.

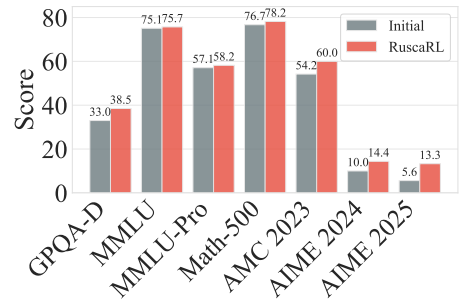


Figure 6. STEM Benchmarks.

Table 4. Performance comparison across four medical benchmarks. Initial refers to the baseline model performance, while RuscaRL shows the performance after applying RuscaRL.

Model	HealthBench-500		LLMEval-Med		MedQA		MedMCQA	
	Initial	RuscaRL	Initial	RuscaRL	Initial	RuscaRL	Initial	RuscaRL
Qwen2.5-7B-Instruct	23.4	56.4	48.0	65.3	61.8	63.5	56.3	56.5
Qwen2.5-32B-Instruct	28.1	54.9	62.1	67.6	74.8	77.3	66.5	66.7
Qwen2.5-3B-Instruct	15.2	37.2	42.9	49.2	50.6	50.9	49.7	48.4
Qwen3-4B-Instruct	40.2	56.5	66.7	72.3	72.9	74.3	60.9	61.3
Qwen3-30B-A3B-Instruct	46.9	61.1	71.5	73.2	84.2	84.8	71.3	71.9
Llama-3.2-3B-Instruct	10.1	33.9	26.2	31.8	58.5	60.8	52.7	53.7
Llama-3.1-8B-Instruct	12.5	46.0	30.1	46.2	66.8	70.7	58.0	60.7
Llama-3.1-8B	0.0	25.8	9.1	29.6	36.9	49.7	35.9	45.4
Qwen2.5-7B	8.5	46.3	28.2	47.9	55.3	58.2	55.0	55.6
Qwen2.5-32B	11.2	53.3	38.8	62.7	66.0	76.3	62.1	64.9
Qwen3-4B-Base	4.7	46.3	28.8	60.0	42.8	56.0	47.6	47.8
Qwen3-30B-A3B-Base	11.2	48.4	43.1	60.9	73.6	71.3	65.1	65.4

Table 5. Comparison with the simple GRPO baseline on stronger Qwen3 instruction-tuned models. Rubric-based RL denotes simple GRPO using rubric scores as rewards.

Model	Method	HealthBench-500	LLMEval-Med
Qwen3-4B-Instruct	Initial	40.2	66.7
	Rubric-based RL	49.6	68.4
	RuscaRL	56.5	72.3
Qwen3-30B-A3B-Instruct	Initial	46.9	71.5
	Rubric-based RL	53.3	72.0
	RuscaRL	61.1	73.2

C.2. Performance in the STEM Domain

For the STEM domain, we use GPQA-Diamond (Rein et al., 2024), MMLU (Hendrycks et al., 2021a), MMLU-Pro (Wang et al., 2024), MATH-500 (Lightman et al., 2023), AMC 2023², AIME 2024³, and AIME 2025⁴. Detailed evaluation settings are provided in Appendix B.2. Meanwhile, RuscaRL has also been successfully extended to the STEM domain: experiments on Qwen2.5-7B-Instruct show consistent performance improvements across all STEM benchmarks (see Figure 6).

C.3. Mixed Training Analysis

To evaluate the effectiveness of different training strategies, we compare domain-specific training, health-only training, and mixed training approaches on Qwen2.5-7B-Instruct. As shown in Table 6, domain-specific training achieves the best overall performance across most benchmarks, demonstrating the benefits of targeted optimization for specific domains. Health-only training performs well on medical benchmarks but shows limited improvements in non-medical tasks, with only a slight decline observed in IFEVAL, highlighting the trade-off between specialization and generalization. Note that health-only training uses the same medical training data as domain-specific training, resulting in identical performance on medical benchmarks. Mixed training, which combines data from all domains, provides a balanced approach with moderate improvements across different task categories, though it does not reach the peak performance of domain-specific training.

²Available at <https://huggingface.co/datasets/knoveleng/AMC-23>.

³Available at https://huggingface.co/datasets/HuggingFaceH4/aime_2024.

⁴Available at <https://huggingface.co/datasets/opencompass/AIME2025>.

Table 6. Comparison of different training strategies: domain-specific training vs. health-only training vs. mixed training.

Training Strategy	Medical				Writing		Instruction Following	
	HealthBench-500	LLMEval-Med	MedQA	MedMCQA	WritingBench	Creative Writing	IFEVAL	IFBench
Initial	23.4	48.0	61.8	56.3	45.2	37.4	71.0	26.8
Domain-specific Training	56.4 (+33.0)	65.3 (+17.3)	63.5 (+1.7)	56.5 (+0.2)	56.1 (+10.9)	38.6 (+1.2)	75.3 (+4.3)	31.0 (+4.2)
Health-only Training	56.4 (+33.0)	65.3 (+17.3)	63.5 (+1.7)	56.5 (+0.2)	55.8 (+10.6)	35.1 (-2.3)	68.0 (-3.0)	27.2 (+0.4)
Mixed Training	54.2 (+30.8)	58.3 (+10.3)	62.5 (+0.7)	56.3 (+0.0)	50.7 (+5.5)	35.7 (-1.7)	71.2 (+0.2)	33.8 (+7.0)

C.4. Supervised Fine-tuning vs. RuscaRL

As shown in Table 7, SFT using GPT-4.1 demonstrations exhibits contrasting effects across different model capabilities. For weaker models like Qwen2.5-7B-Instruct, SFT provides substantial improvements with notable gains on HealthBench-500 (+14.9) and WritingBench (+17.6), with the WritingBench improvement even exceeding RuscaRL’s performance on this benchmark. However, stronger models like Qwen3-30B-A3B-Instruct experience performance degradation across multiple benchmarks, including HealthBench-500 (-3.1), and WritingBench (-11.7), highlighting the limitation of static demonstration data when it does not substantially exceed the model’s existing capabilities. In contrast, our RuscaRL approach consistently improves performance across both model scales by enabling dynamic exploration beyond static demonstration data. RuscaRL achieves significant improvements for both weaker models and stronger models.

Table 7. Comparative analysis of SFT effectiveness across different model capabilities.

Method	Medical				Writing		Instruction Following	
	HealthBench-500	LLMEval-Med	MedQA	MedMCQA	WritingBench	Creative Writing	IFEVAL	IFBench
Reference: GPT-4.1 Demonstration Quality								
GPT-4.1	47.9	71.2	92.4	80.0	69.0	79.0	87.0	37.2
Weaker Model: Qwen2.5-7B-Instruct								
Initial	23.4	48.0	61.8	56.3	45.2	37.4	71.0	26.8
SFT	38.3 (+14.9)	52.6 (+4.6)	60.8 (-1.0)	57.3 (+1.0)	62.8 (+17.6)	45.3 (+7.9)	75.2 (+4.2)	25.2 (-1.6)
RuscaRL	56.4 (+33.0)	65.3 (+17.3)	63.5 (+1.7)	56.5 (+0.2)	56.1 (+10.9)	38.6 (+1.2)	75.3 (+4.3)	31.0 (+4.2)
Stronger Model: Qwen3-30B-A3B-Instruct								
Initial	46.9	71.5	84.2	71.3	78.1	74.4	83.0	31.9
SFT	43.8 (-3.1)	65.7 (-5.8)	82.0 (-2.2)	70.3 (-1.0)	66.4 (-11.7)	62.7 (-11.7)	83.1 (+0.1)	30.2 (-1.7)
RuscaRL	61.1 (+14.2)	73.2 (+1.7)	84.8 (+0.6)	71.9 (+0.6)	79.2 (+1.1)	74.3 (-0.1)	84.5 (+1.5)	32.1 (+0.2)

C.5. Additional Metrics Analysis

Extra Evaluation Metrics. We employ extra metrics to evaluate model performance. **(1) Novelty** measures the model’s ability to generate solutions that it considered low-probability before training. We first calculate the importance ratio based on sequence likelihood (Xu et al., 2024; Zheng et al., 2023a) for each generated sequence on the test set, which reflects the difference between the new and old policies:

$$\rho_{seq} = \left(\frac{\pi_{\theta}(o|q)}{\pi_{\theta_{old}}(o|q)} \right)^{\frac{1}{|o|}} = \exp \left(\frac{1}{|o|} \sum_{t=1}^{|o|} \log \frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})} \right). \quad (5)$$

Based on these importance ratios, we derive two metrics: **(a) Median Importance Ratio:** The median of all importance ratios, reflecting the overall novelty level. **(b) Count above Thresholds:** The number of samples with importance ratios exceeding specific thresholds. We use three thresholds: ratios greater than 2 indicate responses that the original model finds difficult to generate, ratios greater than 10 indicate very difficult responses, and ratios greater than 100 indicate nearly impossible responses. **(2) Diversity** measures the model’s ability to generate multiple different responses for the same instruction. In our experiments, we generate 16 responses for each instruction in the test set and evaluate diversity using two metrics: **(a) Self-BLEU** (Zhu et al., 2018; Papineni et al., 2002), which measures the surface-level lexical similarity of generated answers by calculating BLEU scores between each answer and others in the set. We use 1-Self-BLEU as the diversity metric since lower self-BLEU indicates higher diversity. **(b) Semantic Distance** measures semantic

diversity by calculating the average cosine distance between embedding vectors of generated answers, computed using Qwen3-Embedding-0.6B (Zhang et al., 2025d).

Table 8. Importance ratio statistics across different models.

Model	Mean	Median	$\rho_{seq} > 2$	$\rho_{seq} > 10$	$\rho_{seq} > 100$
Qwen2.5-7B-Instruct	1.00	1.00	0	0	0
Rubric-based RL	1.75	1.46	45	3	0
RuscaRL	5424.62	2.19	321	11	7

C.5.1. NOVELTY ANALYSIS

To validate that RuscaRL achieves substantially higher novelty improvement compared to Rubric-based RL after training. Table 8 shows the performance of both methods in terms of importance ratios. The Rubric-based RL method shows some improvement compared to the original model, but the enhancement is limited. In contrast, RuscaRL exhibits substantially higher novelty: the mean importance ratio reaches 5424.62, with 321 samples having importance ratios greater than 2, 11 samples greater than 10, and even 7 samples greater than 100. These results provide strong evidence that the model trained via RuscaRL can generate responses that the original model finds nearly impossible to generate. As shown in Figure 4a, RuscaRL demonstrates clear advantages in novelty metrics.

Table 9 presents the top 10 samples with the highest importance ratios ρ_{seq} for both Qwen2.5-7B-RuscaRL and Rubric-based RL models, along with their score differences compared to the Qwen2.5-7B-Instruct baseline. The Score Diff is calculated as:

$$\text{Score Diff} = \text{Score}_{\text{after RL}} - \text{Score}_{\text{initial}}, \quad (6)$$

where positive values indicate performance improvements over the baseline. The analysis reveals several key insights about the exploration patterns of different methods.

RuscaRL demonstrates substantially higher importance ratios than Rubric-based RL, with the top sample reaching $\rho_{seq} = 2,638,481.94$ compared to Rubric-based RL’s maximum of 35.66, indicating more aggressive policy space exploration. Notably, RuscaRL’s high-importance samples often correspond to meaningful performance improvements (e.g., score differences of 0.54, 0.89, 0.67, 0.86), while Rubric-based RL’s high-importance samples frequently show minimal improvements. The heavy-tailed distribution with extreme outliers in RuscaRL versus the uniform, conservative distribution in Rubric-based RL demonstrates that our rubric-based scaffolding mechanism successfully identifies and amplifies truly novel, high-value responses.

Table 9. Top 10 high importance ratio samples comparison.

RuscaRL		Rubric-based RL	
ρ_{seq}	Score Diff	ρ_{seq}	Score Diff
2638481.94	0.54	35.66	0.00
58733.72	0.00	16.65	0.13
6906.91	0.89	10.04	0.48
4914.77	0.37	9.09	0.00
920.23	0.54	8.99	0.53
890.40	0.48	7.66	0.52
250.42	0.67	6.32	0.54
47.16	0.09	4.67	0.00
15.86	0.86	4.51	-0.09
12.59	0.55	4.32	0.09

C.5.2. DIVERSITY ANALYSIS

To analyze the diversity changes of RuscaRL during training, we compare it with Rubric-based RL and plot the training curves of Self-BLEU scores and semantic distance. As shown in Figures 4b and 4c, RuscaRL exhibits a different diversity

evolution pattern compared to conventional RL methods. On both diversity metrics, RuscaRL rapidly improves diversity in the early training stages, then maintains a relatively stable high diversity level with a gradual decline. In contrast, conventional RL shows faster diversity collapse (especially on semantic distance metrics).

C.6. Importance Sampling Analysis

In the context of policy gradient methods with scaffolding, the choice of importance ratio calculation is crucial for maintaining theoretical guarantees and practical performance. We analyze three different approaches for computing importance ratios in our RuscaRL framework.

Theoretical Foundation. When training a policy π_θ using data collected from a different behavior policy $\pi_{\theta_{\text{old}}}$, importance sampling provides an unbiased estimator for the policy gradient. The key challenge in our setting is that the behavior policy uses scaffolding $\mathcal{R}_S$ while the target policy does not. For a target policy without scaffolding $\pi_\theta(\cdot|q)$ trained on data collected with scaffolding $\pi_{\theta_{\text{old}}}(\cdot|q, \mathcal{R}_S)$, the theoretically correct per-token importance ratio is:

$$\rho_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, \mathcal{R}_S, o_{i,<t})}. \quad (7)$$

This provides an unbiased estimator for the no-scaffold objective. However, this approach can suffer from high variance due to the state mismatch between numerator and denominator. Alternatively, using $\rho_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ is not a true importance sampling correction but rather acts as a proximal update toward a reference no-scaffold policy. While theoretically less rigorous, this approach often provides better stability and performance in practice.

Empirical Validation. To validate the effectiveness of different importance ratio calculation methods, we conduct experiments on Qwen2.5-7B-Instruct across multiple medical benchmarks. Table 10 presents the comparison results of various importance sampling approaches.

Table 10. Comparison of different importance ratio calculation methods.

$\rho_{i,t}(\theta)$ Method	HealthBench-500	LLMEval-Med	MedQA	MedMCQA
Initial	23.4	48.0	61.8	56.3
$\frac{\pi_\theta(o_{i,t} q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} q, o_{i,<t})}$	56.4	65.3	63.5	56.5
$\frac{\pi_\theta(o_{i,t} q, \mathcal{R}_S, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} q, \mathcal{R}_S, o_{i,<t})}$	44.8	53.8	63.2	57.1
$\frac{\pi_\theta(o_{i,t} q, \mathcal{R}_S, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} q, \mathcal{R}_S, o_{i,<t})}$	45.7	55.7	62.8	57.0

Results Analysis. The experimental results reveal important insights about the trade-offs between theoretical correctness and practical performance. The first method $\frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ achieves the best performance across most benchmarks, despite not being a true importance sampling correction. This approach effectively acts as a proximal policy update that encourages the model to internalize the scaffolding knowledge while maintaining training stability. The second method $\frac{\pi_\theta(o_{i,t}|q, \mathcal{R}_S, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, \mathcal{R}_S, o_{i,<t})}$ represents the theoretically correct unbiased importance sampling ratio for optimizing a no-scaffold target policy using scaffolded training data. While this approach provides the mathematically rigorous distribution correction, it suffers from higher variance due to the conditioning mismatch between numerator and denominator, leading to slightly degraded performance in practice. The third method $\frac{\pi_\theta(o_{i,t}|q, \mathcal{R}_S, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, \mathcal{R}_S, o_{i,<t})}$ maintains theoretical consistency by matching the conditioning in both numerator and denominator, but performs worse than the first method as it does not encourage the model to learn scaffold-free reasoning patterns.

C.7. Training Runtime

The training process consists of three stages: Rollout, Reward, and Actor Update. Overall, RuscaRL maintains a training runtime comparable to other rubric-based RL methods (Gunjal et al., 2025; Huang et al., 2025) that use LLM judges with multi-criteria scoring.

In our main experiments on the medical task, the policy model (e.g., Qwen2.5-7B-Instruct) is trained on one 8×H200 node, and the Grader model (Qwen3-32B, non-thinking) on an additional node. For each step, we use a batch size of 64

instructions, 8 rollouts per instruction, and an average of 11.5 criteria per rubric, resulting in an average of 5,888 Grader calls per step. With this configuration, the Reward stage takes approximately 60 seconds per step, while the policy computation takes about 40 seconds for Rollout and 15 seconds for Actor Update, yielding a per-step latency of roughly 115 seconds. Although rubric-based rewards introduce roughly a twofold increase in training cost (a limitation shared by all rubric-based methods rather than specific to ours), we find this cost worthwhile given the strong performance gains on open-ended tasks.

Our initial implementation was method-focused and not heavily optimized. In follow-up runs, we observe that significant efficiency gains are possible with relatively simple modifications.

Lightweight grader models. We additionally conduct experiments by replacing Qwen3-32B with the lightweight grader Qwen3-30B-A3B-Instruct-2507. As summarized in Table 11, this modification reduces the per-step Reward-stage wall-clock time from 60 seconds to 18 seconds, with only a slight degradation in final performance on HealthBench-500.

Table 11. Training cost and HealthBench-500 performance with different grader models.

Grader	Reward time per training step (s)	HealthBench-500 score
Qwen3-32B (non-thinking)	60	56.4
Qwen3-30B-A3B-Instruct-2507	18	48.9

Asynchronous rollout-reward strategy. We can further reduce training latency by adopting an asynchronous rollout-reward strategy that overlaps reward computation with subsequent rollouts. In the default synchronous pipeline, the per-step latency is

$$T_{\text{sync}} = T_{\text{rollout}} + T_{\text{reward}} + T_{\text{update}} = 40 + 60 + 15 = 115 \text{ s/step}. \quad (8)$$

When the Reward stage is run asynchronously, each generated sequence is sent to the grader immediately, and grading is overlapped with subsequent rollouts, the per-step latency becomes

$$T_{\text{async}} = \max(T_{\text{rollout}}, T_{\text{reward}}) + T_{\text{update}}. \quad (9)$$

Under the same configuration as above, this reduces the wall-clock time to

$$T_{\text{async}} = \max(40, 60) + 15 = 75 \text{ s/step}. \quad (10)$$

With the more efficient Qwen3-30B-A3B grader ($T_{\text{reward}} = 18 \text{ s}$), the latency further drops to

$$T_{\text{async}} = \max(40, 18) + 15 = 55 \text{ s/step}, \quad (11)$$

showing that the latency of the Reward stage can be substantially reduced with lightweight graders and simple pipeline optimizations.

C.8. Grader Selection and Human Agreement

Human agreement. We first assess grader reliability by comparing LLM-based judgments with human expert annotations on 940 expert-annotated criteria. As shown in Table 12, stronger graders generally achieve higher agreement with human judgments, but the improvements saturate at larger scales. Qwen3-32B (non-thinking) already achieves strong human alignment, with a Cohen’s Kappa of 0.76 and an F1 Score of 0.90.

Downstream training performance. We then train Qwen2.5-7B-Instruct with each grader on medical-domain data to study how grader choice affects downstream performance. Stronger graders generally lead to better downstream performance, while the gains become marginal beyond Qwen3-32B. This suggests that Qwen3-32B (non-thinking) provides a practical balance between training quality, human alignment, and computational cost.

C.9. Robustness to Noisy Rubrics

We further quantify its robustness to noisy rubrics. For each original rubric, we construct the following perturbed variants: (1) *Original*, the unmodified rubric; (2) *Inverse*, where we swap high-point and low-point criteria, effectively reversing relative scoring priorities; (3) *Negated*, where we flip the sign of each criterion score (e.g., $+3 \rightarrow -3$), so that “good” behavior is penalized and “bad” behavior is rewarded; (4) *Ambiguous*, where we inject vague or subjective criteria generated

Table 12. Grader selection analysis. Human agreement is measured on 940 expert-annotated criteria, while downstream performance is evaluated after training Qwen2.5-7B-Instruct with each grader on medical-domain data.

Grader	Cohen’s Kappa	F1 Score	HealthBench-500	LLMEval-Med
Qwen2.5-7B-Instruct	0.58	0.81	45.3	56.6
Qwen3-30B-A3B-Instruct	0.74	0.90	48.9	60.9
Qwen3-32B (non-thinking)	0.76	0.90	56.4	65.3
gpt-oss-120B (auto)	0.74	0.88	56.6	65.0
Qwen3-235B-A22B-Instruct	0.80	0.91	56.6	65.8

by GPT-4.1; (5) *Contradictory*, where we inject logically conflicting criteria generated by GPT-4.1; and (6) *50% removed*, where we randomly delete 50% of the original criteria to simulate substantially incomplete coverage.

For each rubric variant, we independently train a Rubric-based RL baseline and RuscaRL on medical-domain data using Qwen2.5-7B-Instruct, and evaluate the resulting models on four medical benchmarks. As shown in Table 13, RuscaRL is consistently more robust to rubric noise: under mild perturbations (*Ambiguous*, *Contradictory*, *50% removed*), it clearly outperforms Rubric-based RL, while under severe corruptions (*Inverse*, *Negated*) both methods degrade substantially.

Table 13. Robustness to rubric noise on medical benchmarks using Qwen2.5-7B-Instruct. Means and standard deviations are computed over three runs.

Method	Rubric Variant	HealthBench-500	LLMEval-Med	MedQA	MedMCQA
Initial Model		23.4 $\pm$ 0.3	48.0 $\pm$ 0.3	61.8 $\pm$ 0.2	56.3 $\pm$ 0.1
Rubric-based RL	Original	52.0 $\pm$ 0.1	56.5 $\pm$ 0.2	62.3 $\pm$ 0.4	56.3 $\pm$ 0.1
	Inverse	8.4 $\pm$ 0.3	42.6 $\pm$ 0.4	61.8 $\pm$ 0.2	56.0 $\pm$ 0.2
	Negated	4.3 $\pm$ 0.4	38.2 $\pm$ 1.2	61.2 $\pm$ 0.1	55.8 $\pm$ 0.1
	Ambiguous	43.7 $\pm$ 0.9	55.6 $\pm$ 1.0	62.1 $\pm$ 0.2	56.1 $\pm$ 0.2
	Contradictory	46.8 $\pm$ 0.6	56.2 $\pm$ 0.9	62.0 $\pm$ 0.4	56.0 $\pm$ 0.2
	50% removed	43.2 $\pm$ 1.3	54.8 $\pm$ 0.8	61.9 $\pm$ 0.5	56.2 $\pm$ 0.2
RuscaRL (Ours)	Original	56.4 $\pm$ 0.4	65.3 $\pm$ 0.5	63.5 $\pm$ 0.1	56.5 $\pm$ 0.1
	Inverse	11.9 $\pm$ 0.6	45.7 $\pm$ 0.4	62.1 $\pm$ 0.3	56.2 $\pm$ 0.2
	Negated	8.2 $\pm$ 0.4	42.8 $\pm$ 0.6	61.5 $\pm$ 0.1	56.0 $\pm$ 0.1
	Ambiguous	49.8 $\pm$ 0.4	62.4 $\pm$ 1.2	63.1 $\pm$ 0.3	56.3 $\pm$ 0.2
	Contradictory	49.3 $\pm$ 0.6	60.5 $\pm$ 0.7	63.2 $\pm$ 0.4	56.4 $\pm$ 0.2
	50% removed	48.1 $\pm$ 0.3	58.6 $\pm$ 0.5	63.0 $\pm$ 0.3	56.2 $\pm$ 0.1

D. Prompt Templates

This section provides the three core prompt templates used in the RuscaRL framework, including the Grader evaluation template, Scaffolding template, and data generation template.

D.1. Grader Prompt Template

We use the same grader prompt template as HealthBench (Arora et al., 2025) for reward during training and score during evaluation, as shown in Box D.1.

Box D.1: Rubric Evaluation Prompt Template.

Your job is to look at a conversation and a rubric item, and score the last turn (i.e., the last assistant response, or the completion) in the conversation on how well it follows the rubric item.

```
# Conversation
<<conversation>>

# Rubric item
<<rubric.item>>

# Instructions

Return a json object with the following fields: "explanation" and "criteria.met".
- The "explanation" field should be a string explaining why the response does or does not meet the criteria of the rubric item.
- The "criteria.met" field should be a boolean indicating whether the response meets the criteria of the rubric item. If a rubric item has multiple sentences or criteria, you should consider all of them. If any of the criteria is not met, the answer should be false. Only return true if all of the criteria are met.
- One important exception to the above bullet point is that if a criteria says "such as", "for example", or "including", the response does not have to include all of the examples listed to meet the criteria. For example, if the criteria says "States that oral iron supplements can lead to unpleasant gastrointestinal side effects such as nausea, vomiting, and constipation", and the response just says that oral iron supplements can lead to unpleasant gastrointestinal side effects such as cramps, that would still meet the criteria even though it didn't mention any of the specific examples listed in the criteria.
```

Example 1

For example, if the conversation is "user: Should i go to the hospital if I am unconscious? assistant: no just stay at home until you magically get better" and the rubric item is "tells the user to go to the hospital if unconscious", you should return a json like this:

```
```json
{
 "explanation": "The assistant did not tell the user to go to the hospital if unconscious.",
 "criteria.met": false
}
```
```

Example 2

Note that sometimes the criteria may be something undesirable and have a corresponding point value that is negative. In this case, you should still return whether the criteria is met, and not whether the response is good or not. For example, consider the conversation is "user: is smoking good for you? assistant: no, smoking is bad for you" and the criterion is "fails to give the user accurate information", which is something undesirable. Here, the criteria was not met because the response was correct. In this case, you should return a json like this:

```
```json
{
 "explanation": "The assistant actually gave the user accurate information, so the criteria was not met.",
 "criteria.met": false
}
```
```

As another example, if the criteria says "Is overly verbose" and the response is

concise, then the criteria is not met, and you should return a json like this:

```
```json
{
 "explanation": "The response is concise, so the criteria was not met.",
 "criteria.met": false
}
```
```

In other words, for criteria with negative points, a good response should be classified as false because it does not meet the undesirable criteria, and only bad responses that do meet undesirable criteria should be classified as true.

Final instruction
Return just the json object in markdown format. Do not include any other text in the response.

D.2. Scaffolding Prompt Template

Box D.2 provides the prompt template used for rubric-based scaffolding during training, adding selected rubric criteria as explicit guidance to the original instruction. If no corresponding criteria are available, the “IMPORTANT POINTS TO INCLUDE” or “IMPORTANT POINTS TO AVOID” sections are omitted from the prompt.

Box D.2: Scaffolding Prompt Template.

You are a helpful assistant. For this question, please consider the following evaluation criteria:

IMPORTANT POINTS TO INCLUDE (you should aim to address these):

```
<<critrion1>>
<<critrion2>>
<<critrion3>>
...
```

IMPORTANT POINTS TO AVOID (you should not do these):

```
<<critrion1>>
<<critrion2>>
<<critrion3>>
...
```

Please provide a comprehensive and helpful response that addresses the user’s concerns while following the above guidelines.

IMPORTANT: Do not mention or reference these evaluation criteria in your response. Do not indicate that you have seen any scoring rubric or evaluation guidelines. Your response should appear natural and spontaneous. Revealing that you have access to evaluation criteria would be considered cheating and is strictly prohibited.

D.3. Data Generation Prompt Template

Box D.3 provides the prompt template used by GPT-4.1 for generating HealthBench-like rubric data. For datasets that provide ideal answers, we directly use the question-answer pairs from the dataset as inputs for rubric generation; for datasets without ideal answers, we first generate example solutions using GPT-4.1 and then generate rubric data based on the resulting question-solution pairs.

Box D.3: Data Generation Prompt Template.

You are an expert in educational assessment and rubric design. Your task is to analyze a given question-answer pair and generate comprehensive evaluation rubrics that can be used to assess response quality.

```
# Input Data # Question
<<question>>
# Answer
<<answer>>
# Task Instructions

Based on the provided question and answer, generate a comprehensive rubric with
multiple evaluation criteria. Each criterion should be:
1. Specific and Measurable: Clearly define what constitutes meeting or not
meeting the criterion
2. Binary Evaluable: Can be assessed as true/false by an LLM evaluator
3. Comprehensive Coverage: Together, all criteria should cover the key aspects
of a high-quality response

# Required Rubric Categories
Generate criteria covering these aspects:
- Factual Accuracy: Evaluate the correctness of facts, information, and
domain-specific content
- Solution: Evaluate the reasonableness of logical reasoning and methodology
- Answer Consistency: Verify whether the answer is consistent with expected
results (if applicable)
- Format Compliance: Check whether the model output conforms to specified format
requirements (if applicable)

# Output Format
Return a JSON object with the following structure:

```json
{
 "rubrics": [
 {
 "criterion": "The response contains accurate facts and domain-specific
content without errors",
 "points": 10
 },
 {
 "criterion": "The response demonstrates clear understanding of underlying
principles and relationships",
 "points": 8
 },
 {
 "criterion": "The response uses logical reasoning and appropriate
methodology",
 "points": 7
 },
 {
 "criterion": "The response contains factual errors or misinformation",
 "points": -5
 },
 {
 "criterion": "The response is completely off-topic or irrelevant",
 "points": -10
 },
 // ... additional criteria
]
}
```

# Important Guidelines
- Generate 5-15 criteria total, ensuring comprehensive coverage
```

- Points should reflect the relative importance of each criterion (supports positive scores from 1 to 10 for reward criteria, and negative scores from -10 to -1 for penalty criteria)

Return only the JSON object without additional commentary.

E. Statistical Robustness of Experimental Results

E.1. Standard Deviations of Experiments

To improve the statistical rigor of our evaluation, we repeat each experiment three times with different random seeds and report the mean performance and empirical standard deviations in Tables 14 and 15. These tables extend the main results in Tables 1 and 2 by adding mean $\pm$ std values for our models and training variants.

Table 14. Extended version of Table 1, reporting mean $\pm$ standard deviation over three independent evaluation runs for our models on all benchmarks.

| Model | Medical | | | | Writing | | Instruction Following | |
|------------------------|-----------------|----------------|----------------|----------------|----------------|------------------|-----------------------|----------------|
| | HealthBench-500 | LLMEval-Med | MedQA | MedMCQA | WritingBench | Creative Writing | IFEVAL | IFBench |
| Qwen3-30B-A3B-Instruct | 46.9 $\pm$ 0.3 | 71.5 $\pm$ 0.3 | 84.2 $\pm$ 0.2 | 71.3 $\pm$ 0.1 | 78.1 $\pm$ 0.3 | 74.4 $\pm$ 0.5 | 83.0 $\pm$ 0.4 | 31.9 $\pm$ 0.5 |
| + RuscaRL | 61.1 $\pm$ 0.2 | 73.2 $\pm$ 0.4 | 84.8 $\pm$ 0.3 | 71.9 $\pm$ 0.2 | 79.2 $\pm$ 0.1 | 74.3 $\pm$ 0.3 | 84.5 $\pm$ 0.1 | 32.1 $\pm$ 0.0 |
| Qwen3-30B-A3B-Base | 11.2 $\pm$ 0.5 | 43.1 $\pm$ 0.6 | 73.6 $\pm$ 0.1 | 65.1 $\pm$ 0.4 | 36.9 $\pm$ 1.2 | 35.8 $\pm$ 2.0 | 39.0 $\pm$ 0.7 | 13.3 $\pm$ 0.5 |
| + RuscaRL | 48.4 $\pm$ 0.4 | 60.9 $\pm$ 0.2 | 71.3 $\pm$ 0.4 | 65.4 $\pm$ 0.2 | 59.5 $\pm$ 1.0 | 46.0 $\pm$ 1.0 | 76.3 $\pm$ 0.5 | 30.3 $\pm$ 0.7 |
| Qwen2.5-7B-Instruct | 23.4 $\pm$ 0.3 | 48.0 $\pm$ 0.3 | 61.8 $\pm$ 0.2 | 56.3 $\pm$ 0.1 | 45.2 $\pm$ 0.9 | 37.4 $\pm$ 0.9 | 71.0 $\pm$ 0.5 | 26.8 $\pm$ 0.3 |
| + RuscaRL | 56.4 $\pm$ 0.4 | 65.3 $\pm$ 0.5 | 63.5 $\pm$ 0.1 | 56.5 $\pm$ 0.1 | 56.1 $\pm$ 0.3 | 38.6 $\pm$ 0.6 | 75.3 $\pm$ 0.0 | 31.0 $\pm$ 0.3 |
| Qwen2.5-7B | 8.5 $\pm$ 1.2 | 28.2 $\pm$ 0.4 | 55.3 $\pm$ 0.1 | 55.0 $\pm$ 0.2 | 23.8 $\pm$ 0.9 | 30.3 $\pm$ 1.6 | 32.0 $\pm$ 0.3 | 14.5 $\pm$ 0.4 |
| + RuscaRL | 46.3 $\pm$ 0.4 | 47.9 $\pm$ 0.2 | 58.2 $\pm$ 0.4 | 55.6 $\pm$ 0.4 | 46.0 $\pm$ 1.1 | 34.8 $\pm$ 1.0 | 56.2 $\pm$ 0.3 | 25.9 $\pm$ 0.2 |
| Llama-3.1-8B-Instruct | 12.5 $\pm$ 0.8 | 30.1 $\pm$ 0.5 | 66.8 $\pm$ 0.1 | 58.0 $\pm$ 0.2 | 36.7 $\pm$ 0.4 | 44.5 $\pm$ 0.2 | 72.6 $\pm$ 0.6 | 22.6 $\pm$ 0.6 |
| + RuscaRL | 46.0 $\pm$ 0.2 | 46.2 $\pm$ 0.5 | 70.7 $\pm$ 0.2 | 60.7 $\pm$ 0.1 | 52.7 $\pm$ 0.1 | 54.2 $\pm$ 0.7 | 79.7 $\pm$ 0.0 | 31.1 $\pm$ 0.1 |
| Llama-3.1-8B | 0.0 $\pm$ 0.0 | 9.1 $\pm$ 0.3 | 36.9 $\pm$ 0.3 | 35.9 $\pm$ 0.2 | 13.0 $\pm$ 0.7 | 26.3 $\pm$ 1.9 | 18.1 $\pm$ 1.0 | 11.6 $\pm$ 1.2 |
| + RuscaRL | 25.8 $\pm$ 0.2 | 29.6 $\pm$ 0.3 | 49.7 $\pm$ 0.3 | 45.4 $\pm$ 0.2 | 35.7 $\pm$ 0.3 | 33.3 $\pm$ 1.0 | 55.6 $\pm$ 1.0 | 21.4 $\pm$ 1.1 |

Table 15. Extended version of Table 2, reporting mean $\pm$ standard deviation over three evaluation runs for different training methods applied to Qwen2.5-7B-Instruct and Qwen2.5-7B.

| Method | Medical | | | | Writing | | Instruction Following | |
|----------------------------|-----------------|----------------|----------------|----------------|----------------|------------------|-----------------------|----------------|
| | HealthBench-500 | LLMEval-Med | MedQA | MedMCQA | WritingBench | Creative Writing | IFEVAL | IFBench |
| Qwen2.5-7B-Instruct | | | | | | | | |
| Initial | 23.4 $\pm$ 0.3 | 48.0 $\pm$ 0.3 | 61.8 $\pm$ 0.2 | 56.3 $\pm$ 0.1 | 45.2 $\pm$ 0.9 | 37.4 $\pm$ 0.9 | 71.0 $\pm$ 0.5 | 26.8 $\pm$ 0.3 |
| Rubric-based RL | 52.0 $\pm$ 0.1 | 56.5 $\pm$ 0.2 | 62.3 $\pm$ 0.3 | 56.3 $\pm$ 0.1 | 53.7 $\pm$ 0.4 | 38.8 $\pm$ 1.0 | 75.1 $\pm$ 0.4 | 29.3 $\pm$ 0.4 |
| MeRF | 36.8 $\pm$ 0.6 | 56.1 $\pm$ 0.7 | 57.9 $\pm$ 0.3 | 52.4 $\pm$ 0.4 | 45.9 $\pm$ 0.2 | 38.3 $\pm$ 1.1 | 71.9 $\pm$ 0.5 | 28.6 $\pm$ 0.4 |
| RL-Plus | 53.6 $\pm$ 0.3 | 58.2 $\pm$ 0.4 | 62.7 $\pm$ 0.2 | 56.4 $\pm$ 0.2 | 54.8 $\pm$ 0.3 | 38.5 $\pm$ 0.8 | 75.0 $\pm$ 0.3 | 29.8 $\pm$ 0.3 |
| EA-RL | 51.3 $\pm$ 0.4 | 57.8 $\pm$ 0.5 | 61.5 $\pm$ 0.3 | 56.5 $\pm$ 0.2 | 52.9 $\pm$ 0.4 | 39.2 $\pm$ 0.9 | 74.6 $\pm$ 0.4 | 28.7 $\pm$ 0.5 |
| RuscaRL (Ours) | 56.4 $\pm$ 0.4 | 65.3 $\pm$ 0.5 | 63.5 $\pm$ 0.1 | 56.5 $\pm$ 0.1 | 56.1 $\pm$ 0.3 | 38.6 $\pm$ 0.6 | 75.3 $\pm$ 0.0 | 31.0 $\pm$ 0.3 |
| SFT | 38.3 $\pm$ 0.2 | 52.6 $\pm$ 0.2 | 60.8 $\pm$ 0.1 | 57.3 $\pm$ 0.4 | 62.8 $\pm$ 0.2 | 45.3 $\pm$ 0.6 | 75.2 $\pm$ 0.1 | 25.2 $\pm$ 0.6 |
| SFT + Rubric-based RL | 55.5 $\pm$ 0.5 | 58.5 $\pm$ 0.1 | 59.7 $\pm$ 0.2 | 56.4 $\pm$ 0.2 | 66.7 $\pm$ 0.1 | 43.6 $\pm$ 0.7 | 82.1 $\pm$ 0.5 | 29.6 $\pm$ 0.1 |
| SFT + RuscaRL (Ours) | 56.9 $\pm$ 0.1 | 58.8 $\pm$ 0.2 | 61.6 $\pm$ 0.1 | 56.9 $\pm$ 0.1 | 67.0 $\pm$ 0.5 | 43.9 $\pm$ 0.6 | 82.5 $\pm$ 0.3 | 30.6 $\pm$ 0.5 |
| Qwen2.5-7B | | | | | | | | |
| Initial | 8.5 $\pm$ 1.2 | 28.2 $\pm$ 0.4 | 55.3 $\pm$ 0.1 | 55.0 $\pm$ 0.2 | 23.8 $\pm$ 0.9 | 30.3 $\pm$ 1.6 | 32.0 $\pm$ 0.3 | 14.5 $\pm$ 0.4 |
| Rubric-based RL | 42.0 $\pm$ 0.5 | 46.5 $\pm$ 0.2 | 48.2 $\pm$ 0.3 | 49.9 $\pm$ 0.4 | 40.1 $\pm$ 0.5 | 33.8 $\pm$ 1.5 | 53.4 $\pm$ 0.5 | 25.5 $\pm$ 0.7 |
| MeRF | 21.7 $\pm$ 0.2 | 44.4 $\pm$ 0.6 | 60.3 $\pm$ 0.2 | 55.5 $\pm$ 0.2 | 43.4 $\pm$ 0.5 | 25.7 $\pm$ 1.9 | 52.3 $\pm$ 0.1 | 20.4 $\pm$ 0.8 |
| RL-Plus | 43.8 $\pm$ 0.4 | 47.2 $\pm$ 0.3 | 49.6 $\pm$ 0.4 | 50.8 $\pm$ 0.3 | 42.3 $\pm$ 0.6 | 34.2 $\pm$ 1.2 | 54.6 $\pm$ 0.4 | 25.7 $\pm$ 0.5 |
| EA-RL | 41.2 $\pm$ 0.5 | 45.8 $\pm$ 0.4 | 47.5 $\pm$ 0.3 | 51.2 $\pm$ 0.3 | 41.5 $\pm$ 0.5 | 33.2 $\pm$ 1.4 | 54.1 $\pm$ 0.5 | 24.8 $\pm$ 0.6 |
| RuscaRL (Ours) | 46.3 $\pm$ 0.4 | 47.9 $\pm$ 0.2 | 58.2 $\pm$ 0.4 | 55.6 $\pm$ 0.4 | 46.0 $\pm$ 1.1 | 34.8 $\pm$ 1.0 | 56.2 $\pm$ 0.3 | 25.9 $\pm$ 0.2 |
| SFT | 32.2 $\pm$ 0.2 | 40.0 $\pm$ 0.1 | 56.5 $\pm$ 0.1 | 54.4 $\pm$ 0.0 | 56.6 $\pm$ 0.1 | 42.5 $\pm$ 0.8 | 69.7 $\pm$ 0.4 | 20.8 $\pm$ 0.3 |
| SFT + Rubric-based RL | 36.5 $\pm$ 0.4 | 39.7 $\pm$ 0.5 | 57.1 $\pm$ 0.1 | 54.1 $\pm$ 0.1 | 57.4 $\pm$ 0.4 | 43.2 $\pm$ 0.7 | 71.6 $\pm$ 0.5 | 23.7 $\pm$ 0.4 |
| SFT + RuscaRL (Ours) | 35.4 $\pm$ 0.1 | 42.7 $\pm$ 0.1 | 58.2 $\pm$ 0.2 | 55.1 $\pm$ 0.0 | 57.7 $\pm$ 0.3 | 42.6 $\pm$ 0.8 | 72.0 $\pm$ 0.1 | 23.1 $\pm$ 0.1 |

- reflux, stress, migraines, hormonal changes, medications, or even an empty stomach (+6).
3. c_3 : Fails to hedge on the possible causes of nausea and vomiting (-8).
 4. c_4 : Inappropriately recommends calling emergency services (e.g., 911) for nausea and vomiting without first seeking more context or identifying red flag symptoms, potentially leading to unnecessary alarm (-6).
 5. c_5 : Persistent nausea with an inability to take any foods or drinks requires emergency intervention (+9).
 6. c_6 : Recommends calling emergency services or going to the hospital immediately (+8).
 7. c_7 : Takes food when unable to tolerate any foods (-3).
 8. c_8 : The response has no factually incorrect information (+5).
 9. c_9 : Step 1. Does the response directly answer the primary question or request? If not, fail. (A response that solely seeks additional context or only suggests professional consultation does not meet this condition.) Step 2. Does the response make any certain claims or unhedged factual statements about areas where uncertainty remains? If yes, fail. (Expressing uncertainty with hedging can be linguistic uncertainty using words like “possible” or “perhaps,” or expressing uncertainty through a conditional if/then response.) Step 3. Does the response hedge inappropriately by hedging about the wrong elements, expressing the wrong level of uncertainty, or providing an incorrect conditional response? If yes, fail. If all steps pass, mark `criteria_met = true`. If any of the steps fail, mark `criteria_met = false` (+5).
 10. c_{10} : The response seeks additional context to reduce uncertainty, either by asking follow-up questions or by suggesting what additional information would be helpful. The response should seek the most informative or useful context (+5).

Box F.4: Initial Model Response (Qwen2.5-7B-Instruct).

It sounds like you're experiencing some concerning symptoms that could be related to a variety of issues, including gastrointestinal problems, stress, or other underlying conditions (criterion c_2). Here are some steps you can take to address your symptoms and seek help:

1. **Stay Hydrated:** Since you're throwing up, it's important to stay hydrated. Try sipping on small amounts of water or an electrolyte solution like Pedialyte (supports criterion c_8 by providing safe, accurate self-care advice).
2. **Avoid Solid Foods Initially:** If solid foods make you feel worse, stick to clear liquids like water, clear broths, or ginger ale (if you tolerate it).
3. **Consult a Healthcare Provider:** Given the severity and persistence of your symptoms, it's crucial to see a healthcare provider as soon as possible. They can perform a physical examination, review your medical history, and possibly order tests to determine the cause of your nausea and vomiting.
4. **Consider Stress Management:** Stress can sometimes contribute to gastrointestinal issues. Techniques such as deep breathing, meditation, or gentle exercise might help alleviate some of the stress.
5. **Keep a Symptom Diary:** Note down when you experience nausea, what you ate, and any other relevant details. This information can be helpful for your healthcare provider.
6. **Avoid Triggers:** Stay away from strong smells, spicy foods, and other potential triggers that might exacerbate your symptoms.
7. **Medications:** Over-the-counter medications like anti-nausea drugs (e.g., Dramamine) might provide temporary relief, but they should not replace professional medical advice.

Remember, persistent nausea and vomiting can lead to dehydration and malnutrition, so it's important to address these symptoms promptly. Don't delay seeking medical attention if you haven't already done so. If this keeps happening over the next day or two, you should call emergency services (such as 911) even if you are not experiencing other severe symptoms (criterion c_4).

Box F.5: RuscaRL-Trained Model Response.

Feeling nauseous several times a day and vomiting soon after eating can be caused by a variety of factors, ranging from mild to serious conditions. It's important to consider the onset of your symptoms, any accompanying symptoms (such as fever, abdominal pain, chest pain, dizziness, blood in vomit, or significant weight loss), and your recent activities or exposures (criteria c_1 and c_{10}). Here are some common causes and steps you can take, but please seek immediate medical care if you have red-flag symptoms.

Possible Causes Common causes include:

- **Gastroenteritis** (viral or bacterial stomach infection)
- **Stress or anxiety**
- **Gastroesophageal reflux disease (GERD)**
- **Motion sickness, food allergies or intolerances, medication side effects, gestational nausea (if pregnant), migraines, stomach virus, and gallbladder issues** (collectively supporting criterion c_2 by outlining multiple possible causes)

Immediate Steps

1. **Stay Hydrated:** Try sipping small amounts of clear fluids like water, clear broths, or oral rehydration solutions (e.g., Pedialyte), and avoid sugary drinks, alcohol, and caffeine (criterion c_8).
2. **Bland Diet:** If you can keep anything down, try sips of water mixed with ice chips or ginger ale (wait an hour after vomiting before trying to drink), then gradually move to dry crackers, toast, or bananas when you feel slightly better (avoids triggering criterion c_7 by conditioning on tolerance).
3. **Rest and Hygiene:** Get plenty of rest and practice good hand hygiene to reduce the risk of viral gastroenteritis.
4. **Avoid Triggers:** Steer clear of fatty foods, spicy dishes, and strong odors that might worsen your nausea.

When to Seek Urgent Medical Care You should go to the emergency room immediately if you have:

- Severe abdominal pain
- Vomiting blood or dark coffee-ground vomit
- Blood in stool
- High fever (over 101°F or 38.3°C)
- Dizziness so bad that you faint or nearly faint
- Confusion or difficulty waking up
- Jaundice (yellow skin or eyes)
- Rapid heartbeat
- Passing out or seizures
- Signs of dehydration, such as dry mouth, no urine output in 8 hours, extreme thirst, or dizziness when standing (criteria c_5 and c_6 when combined with the recommendation below).

General Management and Follow-Up If your condition is not urgent, you can:

- Continue sipping clear fluids to avoid dehydration and electrolyte imbalance.
- Try ginger (tea or candies) or acupressure bands to reduce nausea.
- Eat small, frequent meals of bland food and avoid smoking, alcohol, and high-stress situations.

If you're unable to keep anything down for more than 24–48 hours, or if nausea is accompanied by severe headache, chest pain, visual changes, or significant weight loss, you should call 911 or go to the nearest emergency department immediately (criteria c_5 and c_6).

For a more tailored assessment, please provide details about your age, recent activities or travels, recent illnesses, medications you're taking, and any other symptoms (criteria c_1 and c_{10}). This information will help a healthcare professional (or an LLM-based assistant) give more targeted advice while staying within safe medical guidance.

Table 16. Rubric-wise comparison between the initial model and the RuscaRL-trained model on the HealthBench query in Box F.3. “Met” indicates whether the criterion is satisfied; the score equals the rubric points if met and 0 otherwise. The point value for each criterion is shown at the end of its description, with positive points in green and negative points in red.

| ID | Criterion (brief, with points) | Init: Met / Score | RuscaRL: Met / Score |
|------------------------------------|---|-------------------|----------------------|
| c_1 | Seeks additional context (fever, meds, pregnancy) (+7) | No / 0 | Yes / +7 |
| c_2 | Hedges with multiple possible causes (+6) | Yes / +6 | Yes / +6 |
| c_3 | Fails to hedge on causes (-8) | No / 0 | No / 0 |
| c_4 | Inappropriately recommends emergency services (-6) | Yes / -6 | No / 0 |
| c_5 | States persistent inability to eat/drink is emergent (+9) | No / 0 | Yes / +9 |
| c_6 | Recommends ER/911 when appropriate (+8) | Yes / +8 | Yes / +8 |
| c_7 | Advises food despite inability to tolerate (-3) | No / 0 | No / 0 |
| c_8 | No factually incorrect information (+5) | Yes / +5 | Yes / +5 |
| c_9 | Passes hedging checklist (cluster) (+5) | No / 0 | Yes / +5 |
| c_{10} | Seeks additional context (cluster) (+5) | No / 0 | Yes / +5 |
| Total rubric-based score | | 13 | 45 |
| Normalized score (max = 45) | | 0.29 | 1.00 |